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Towards understanding energy partition in magnetized collisionless shocks in the laboratory

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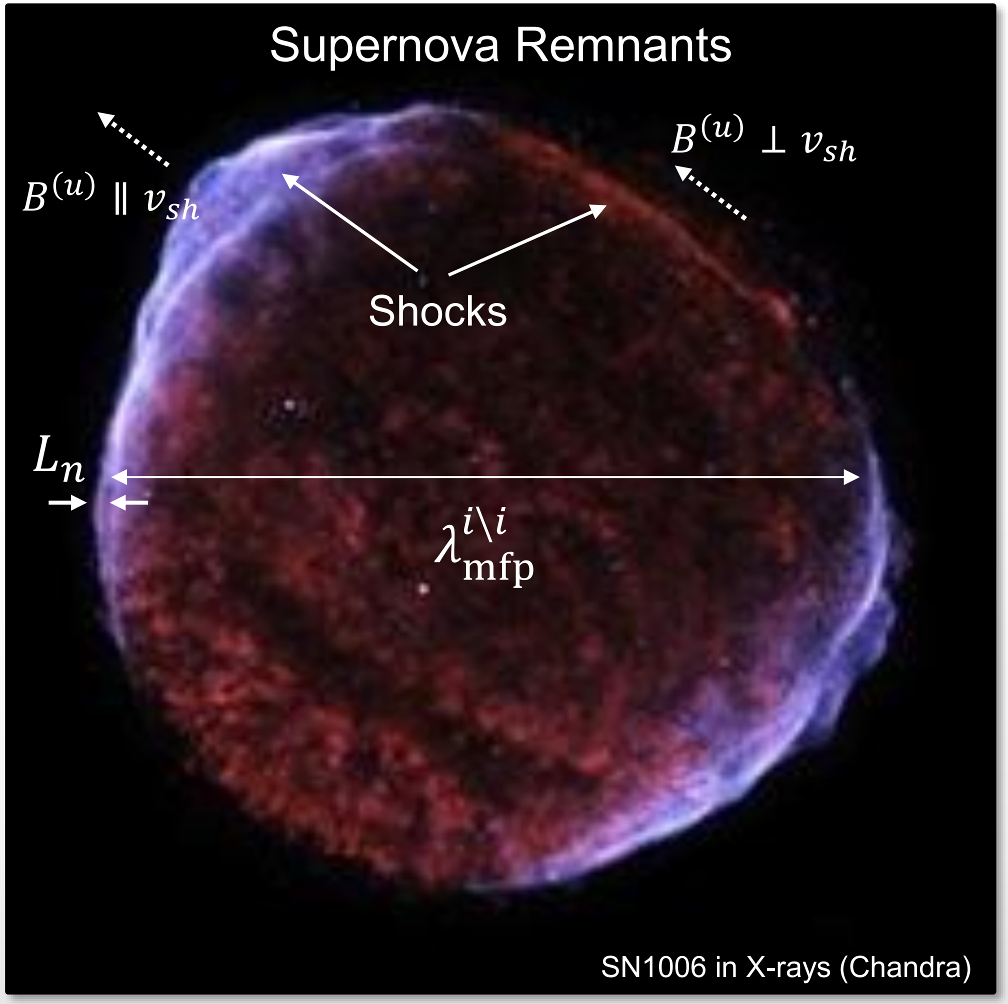
Collisionless shocks in the universe

Supernova remnants are beautiful examples of collisionless shocks.

1. Hierarchy of length-scales

$$\underbrace{L_n}_{\text{Density grad. length-scale}} \ll \underbrace{\lambda_{\mathrm{mfp}}^{i\backslash i}}_{\text{Ion-ion Coulomb mean free path}}$$

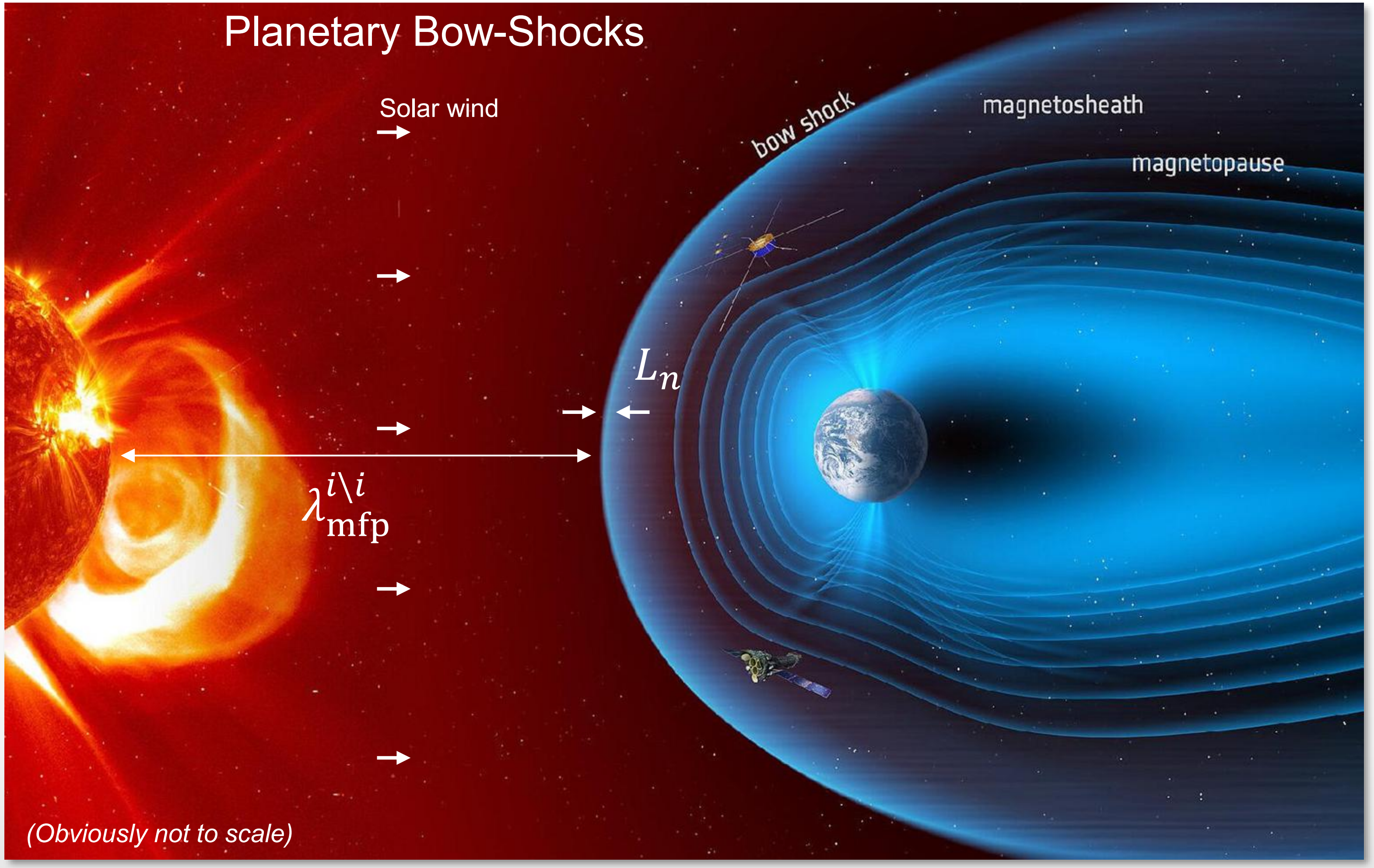
2. Interactions are mediated by EM fields
3. Efficient high-energy particle accelerators
4. High Mach number regime $\mathrm{M_A} > 20$ $(0.2 < \beta < 2)$



Collisionless shocks in the solar system

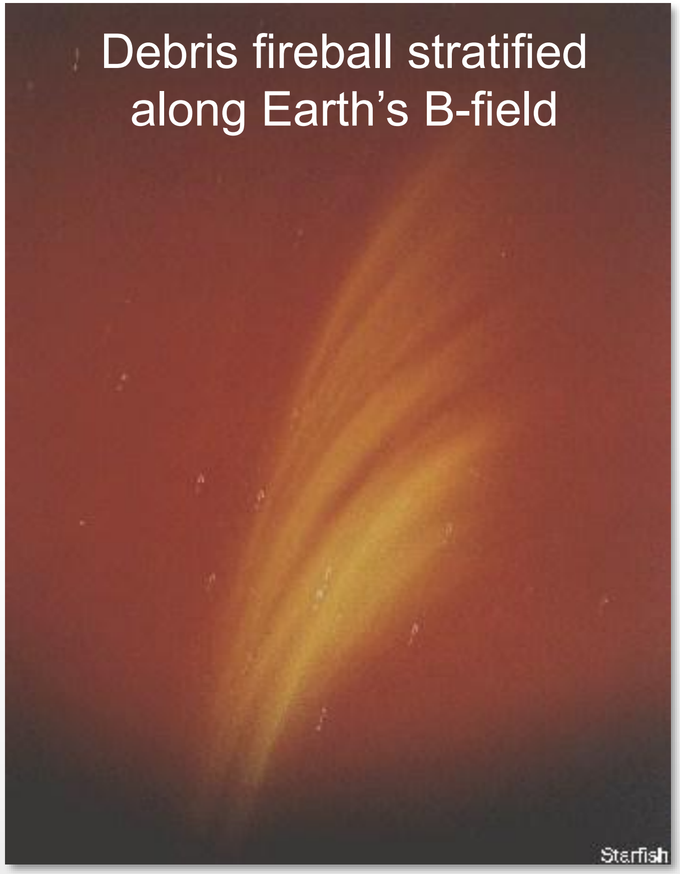
Low/intermediate Mach number regime

$$1 < \mathrm{M_A} < 20$$



Here on Earth, understanding collisionless shocks is crucial for predicting the effect of high-altitude nuclear explosions

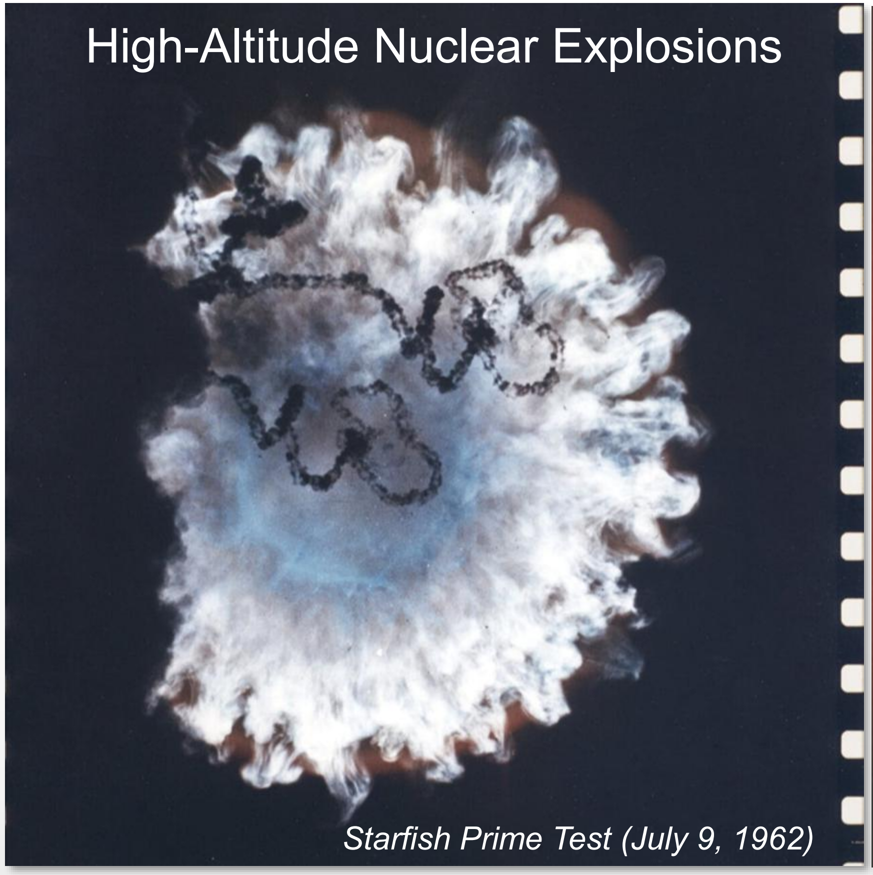
Ground-based photographs of the *Starfish Prime* test from Hawaii

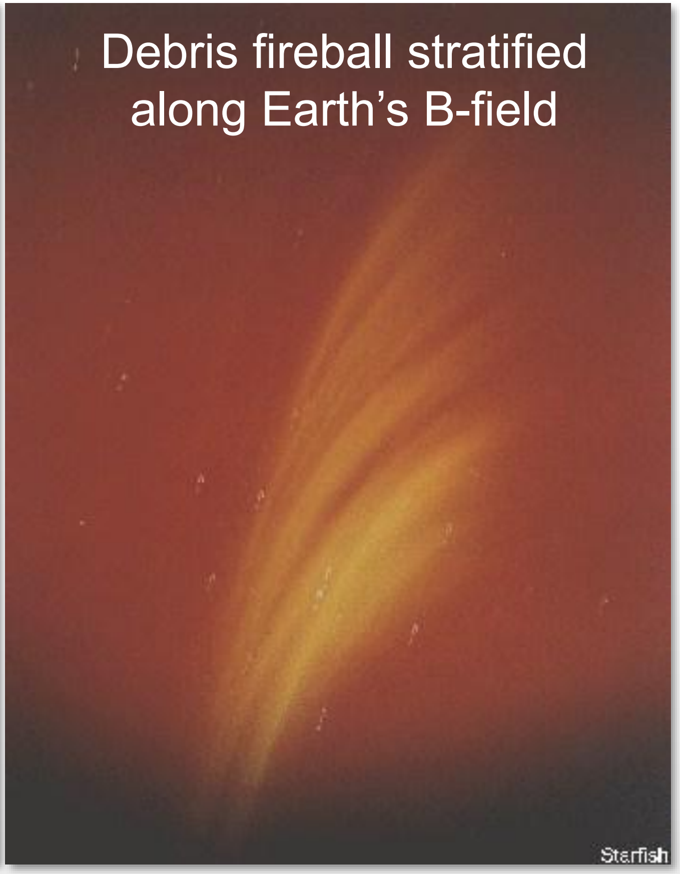


Here on Earth, understanding collisionless shocks is crucial for predicting the effect of high-altitude nuclear explosions

HANEs are of critical interest to security:

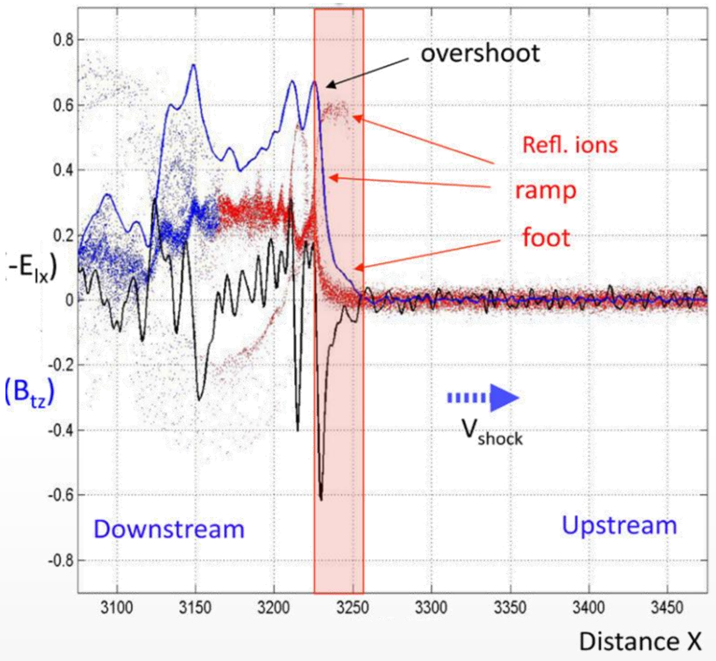
1. Nuclear radiation **creates energetic electrons**
2. Mass and momentum transport **affected by Earth's magnetic field**
3. Exo-atmospheric collisionless shocks can **accelerate charged particles**
4. **May endanger satellites**, critical infrastructure in the modern era





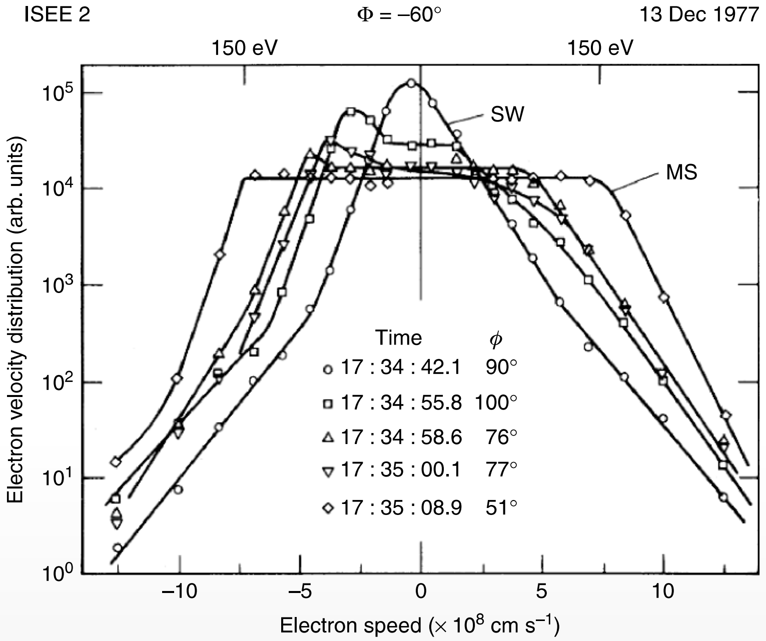
Energy partitioning in collisionless shocks is an outstanding problem

Microphysics of collisionless shocks



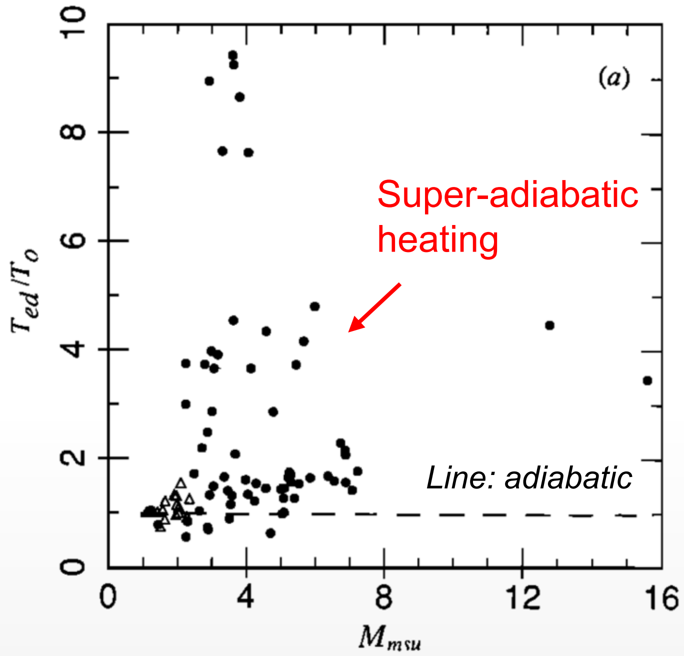
- Reflected ions convert ram to thermal pressure
- Primarily heated perp to B_0

Non-Maxwellian VDFs



- Small fraction (~10%) of total energy compared to ions
- "Flat-top" distributions observed

Electron heating vs Alfvenic Mach number

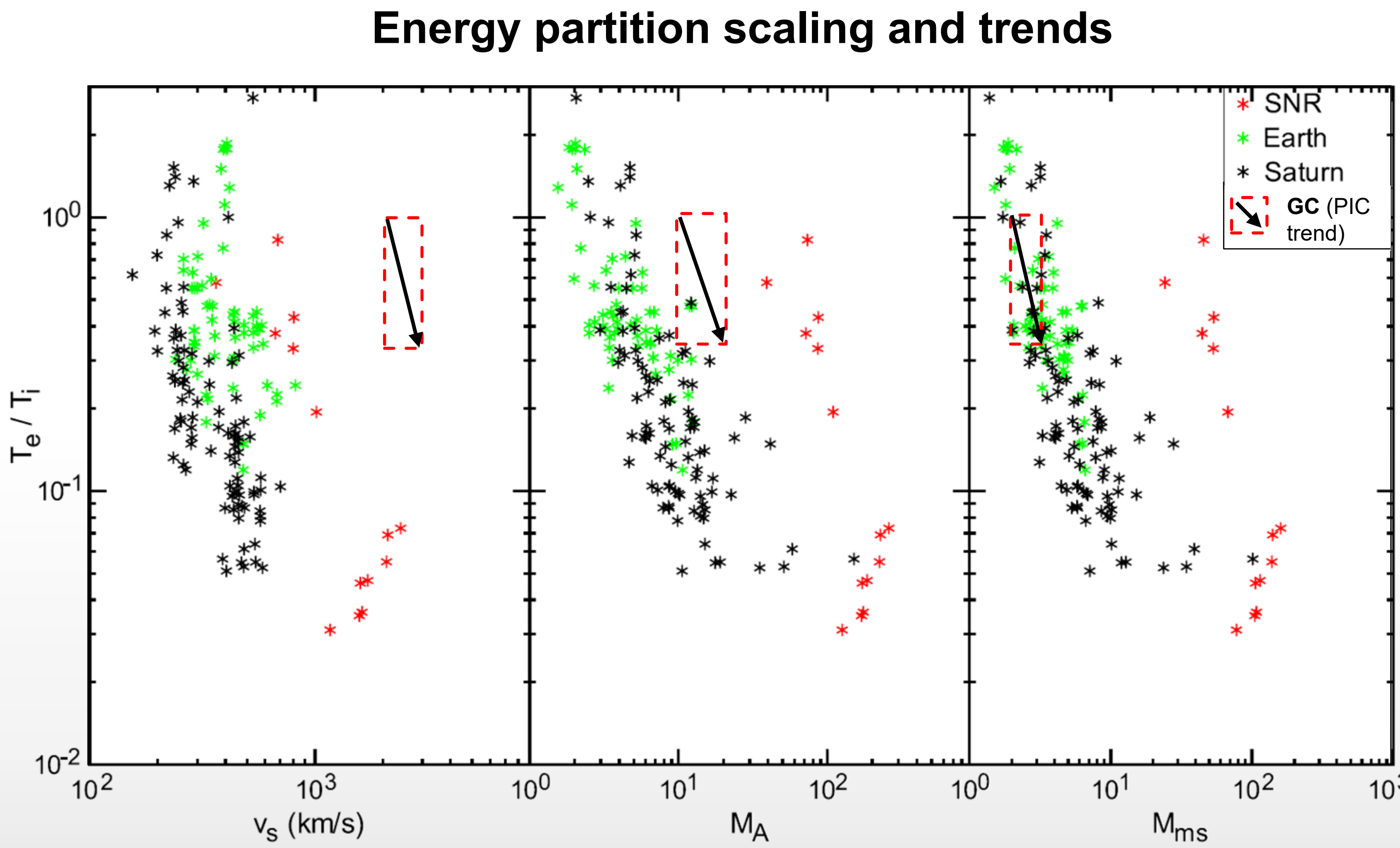


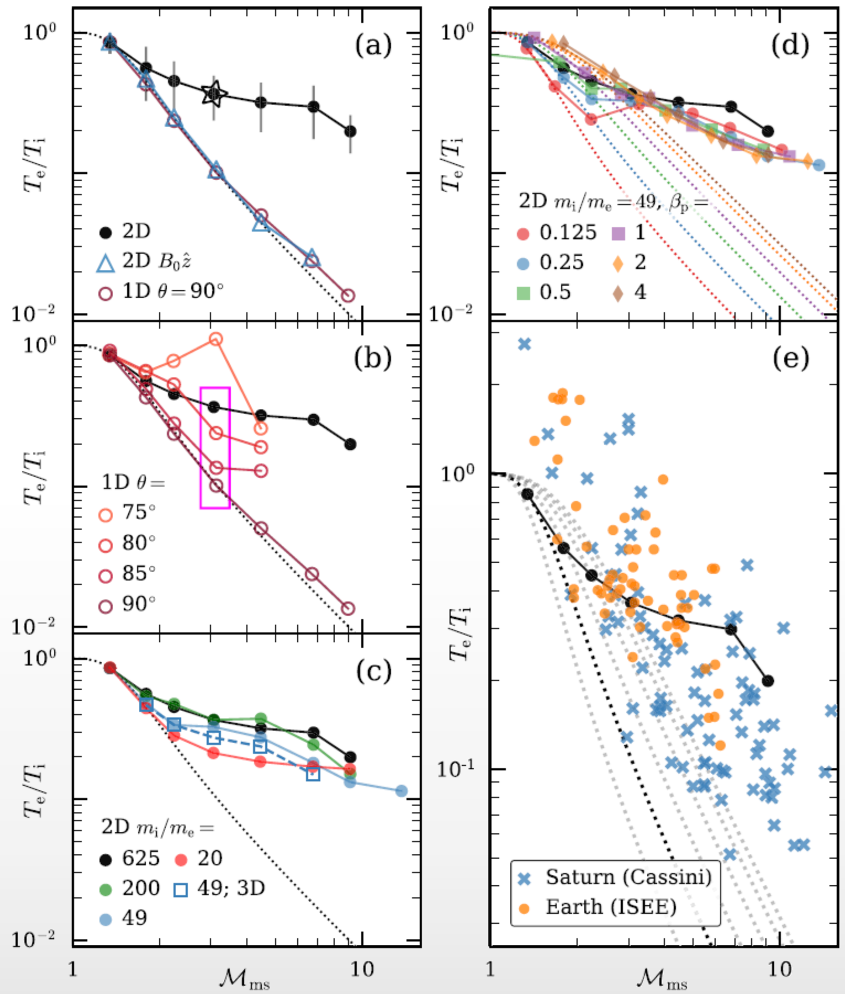
- Significant non-adiabatic heating observed at higher M_A
- Potentially non-isotropic heating

Laboratory experiments coupled with numerical simulations can help address this question

Feldman et al., J. Geophys. Res. (1983)
Schwartz et al., J. Geophys. Res. (1988)
Marcowith et al., Rep. Prof. Phys. (2016)

Simulations indicate $T_e/T_i \to 0.1$ as $M_{ms} \to 10$, but data shows large statistical variance

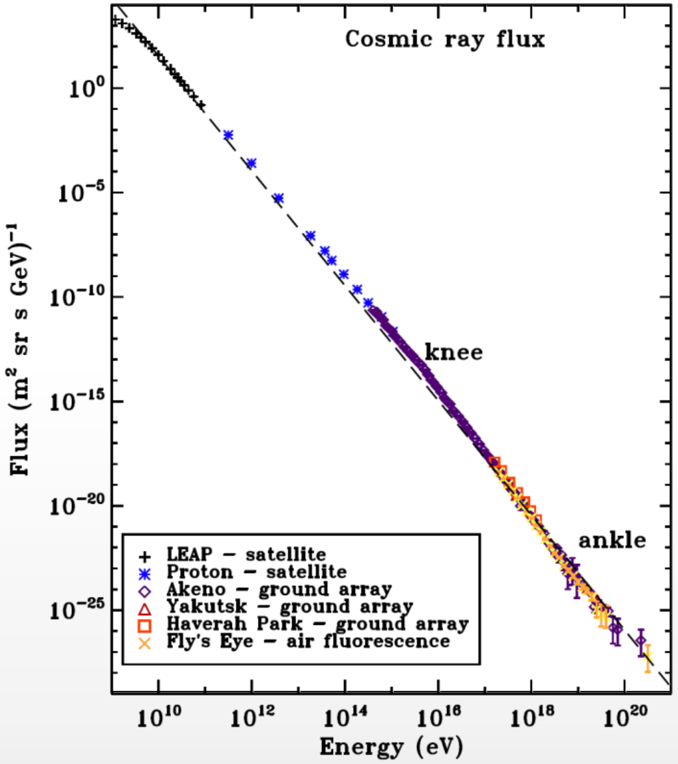


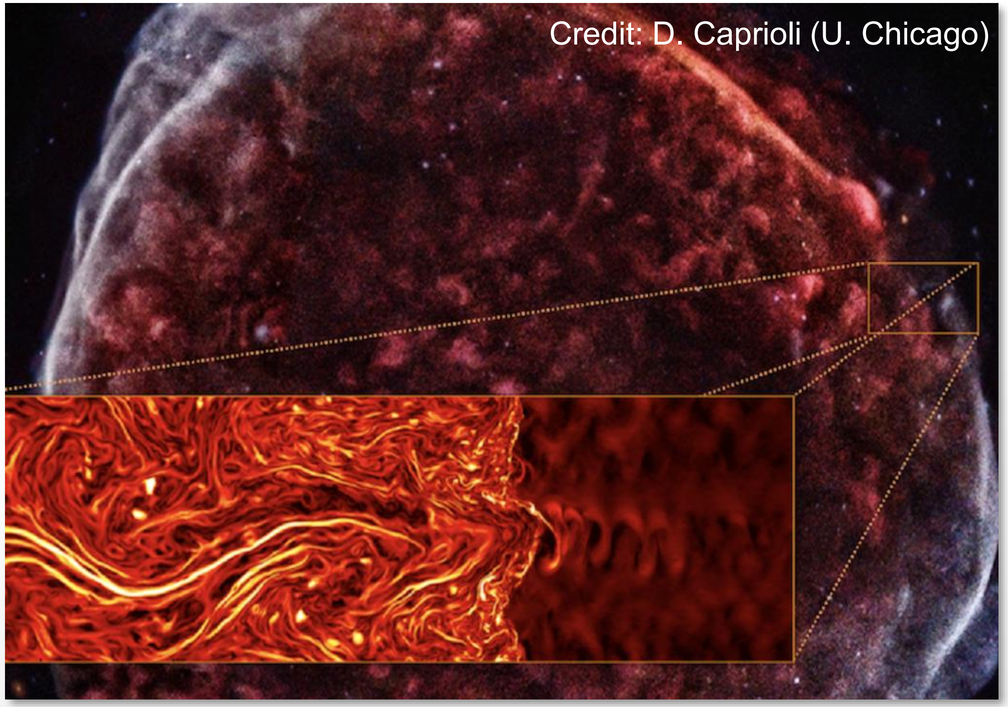


Ghavamian et al. Sp. Sci. Rev (2013)
Chen et al., Phys. Rev. Lett. (2018)
Tran & Sironi, Astrophys. J. (2020)
Raymond et al., Astrophys. J. (2023)

Particle acceleration in collisionless shocks

- **The galaxy is filled with energetic particles** called cosmic rays
- It is believed that **supernova remnants** are sites of efficient particle acceleration





Helder et al., Sp. Sci. Rev. (2013)
Caprioli & Spitkovsky, Astroph. J. (2014)

Particle acceleration in collisionless shocks

Ion acceleration in a collisionless shock:

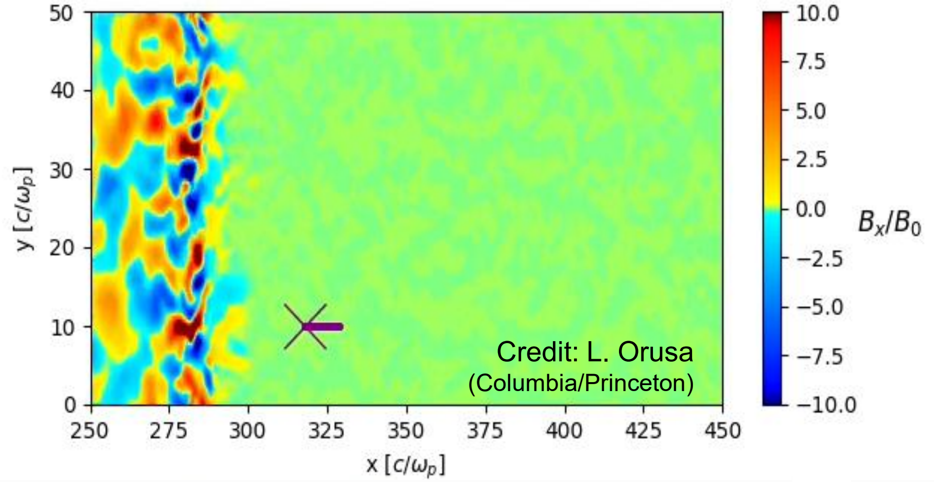
1. **Ions are reflected** by the shock into the upstream

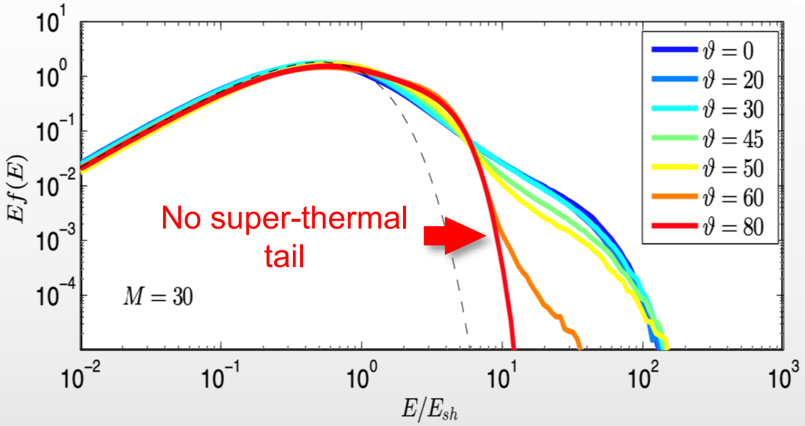
2. The **ions gyrate back and enter the downstream** where they scatter with plasma waves/turbulence

3. If the magnetic field allows, **ions can cross back into the fluctuating upstream**, and the energization continues

4. It is *believed* (foreshadowing) that **perpendicular shocks** $80^\circ \leq \vartheta \leq 90^\circ$ **do not allow multiple crossings**, and can't accelerate ions

Ion energization in a magnetized shock



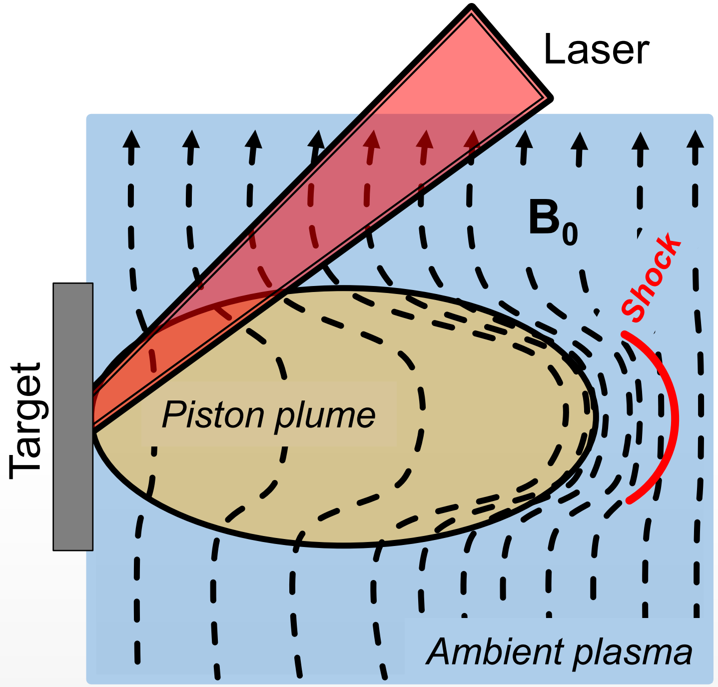


Caprioli & Spitkovsky, Astroph. J. (2014)
Orusa et al., (in review at Astrophys. J.)

Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

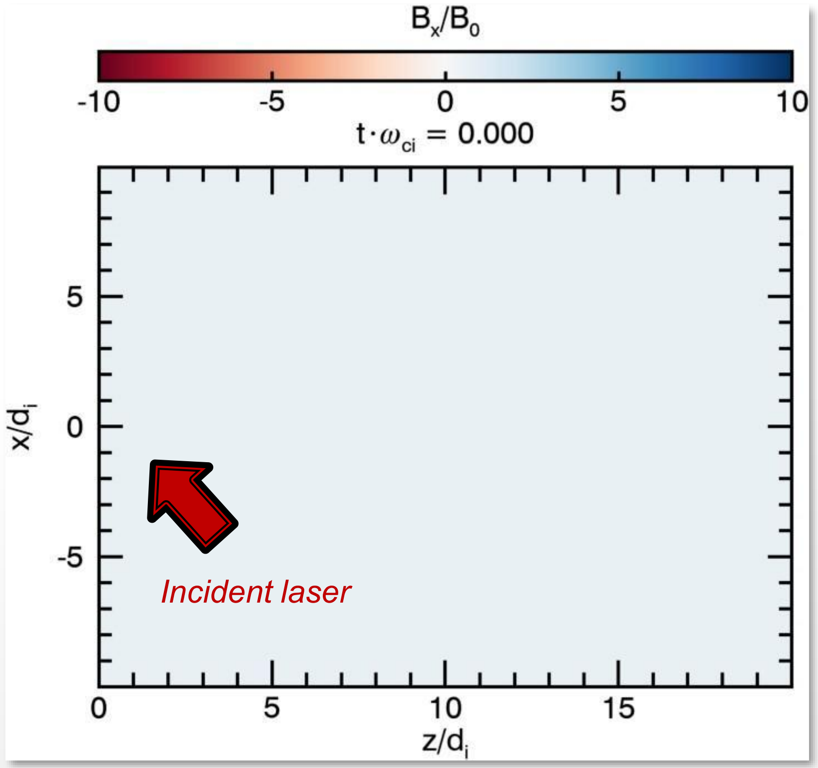
Laboratory experiments use fast laser-driven piston plasmas to generate magnetized collisionless shocks

Piston-driven shock formation



- High-powered lasers generate hypersonic piston plasma
- The piston expands through magnetized ambient plasma to drive collisionless shock

Simulation of collisionless shock formation in the laboratory



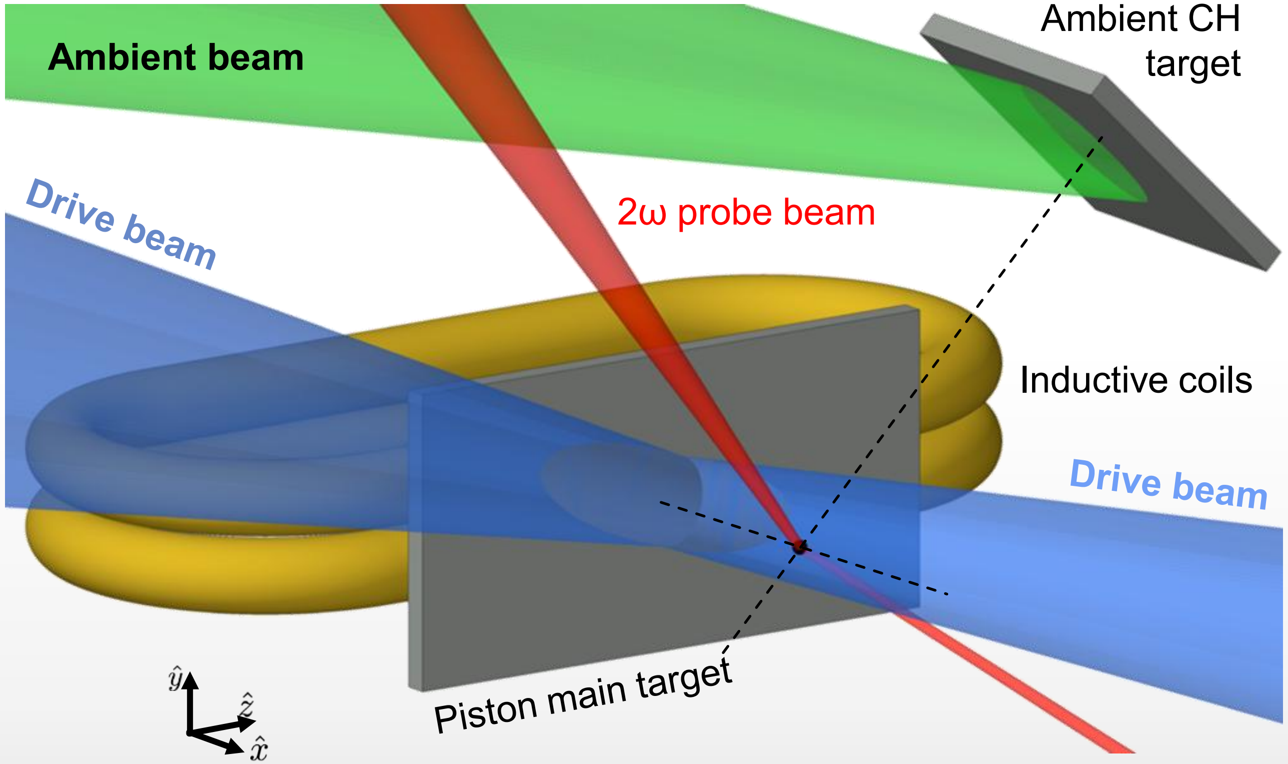
- 2D particle-in-cell simulation with the code PSC

Germaschewski et al., J. Comp. Phys. (2016)
Bodarenko et al., Nat. Phys. (2017)

Fox et al., Phys. Plasmas (2018)

Previous experiments have demonstrated the formation of collisionless shock *precursors*

Experimental setup



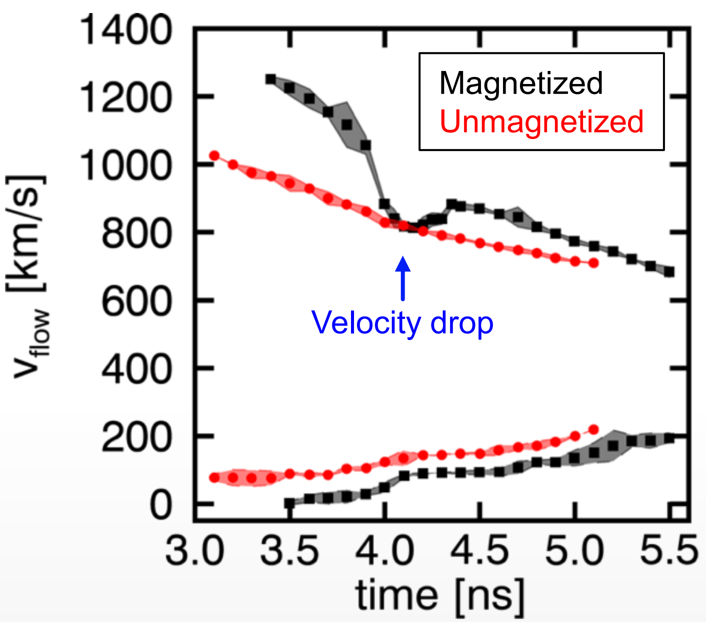
Parameters

Upstream B-field = 10 T
Upstream density = 2e18 cm^{-3}
Shock velocity ~ 750 km/s
M_A ~ 15
M_S ~ 15
Probe duration = 2 ns
Probe at $\approx 30\ d_i$
Ion gyrations at probe = 2

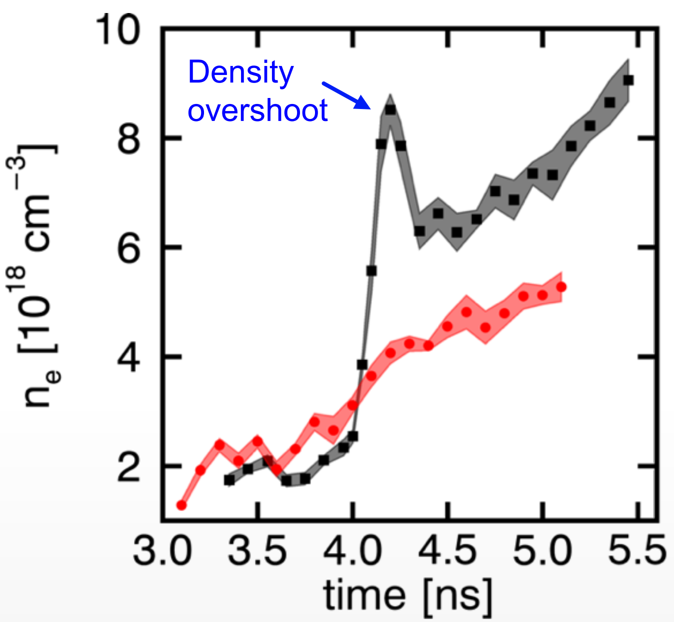
Schaeffer et al., Phys. Rev. Lett. (2017)
Schaeffer et al., Phys. Rev. Lett. (2019)

Previous experiments have demonstrated the formation of collisionless shock *precursors*

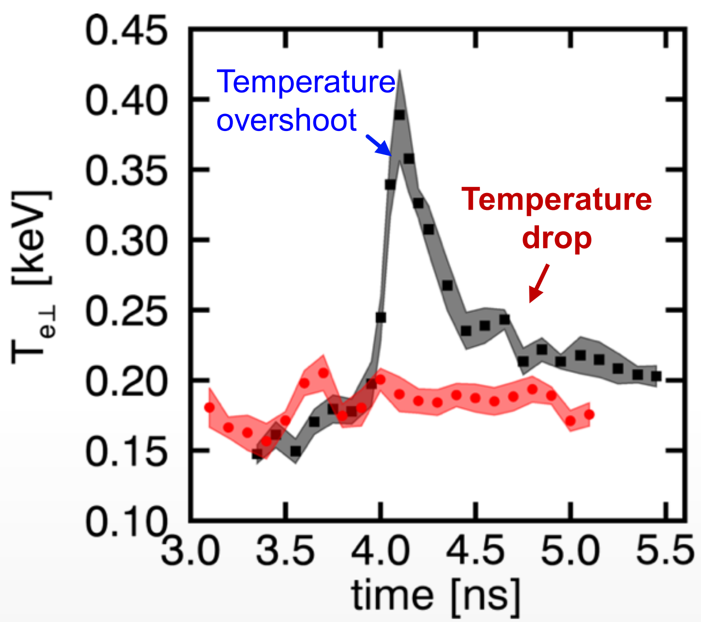
Ion flow speed



Electron density



Electron temperature



Decelerated piston flow and accelerated ambient flow

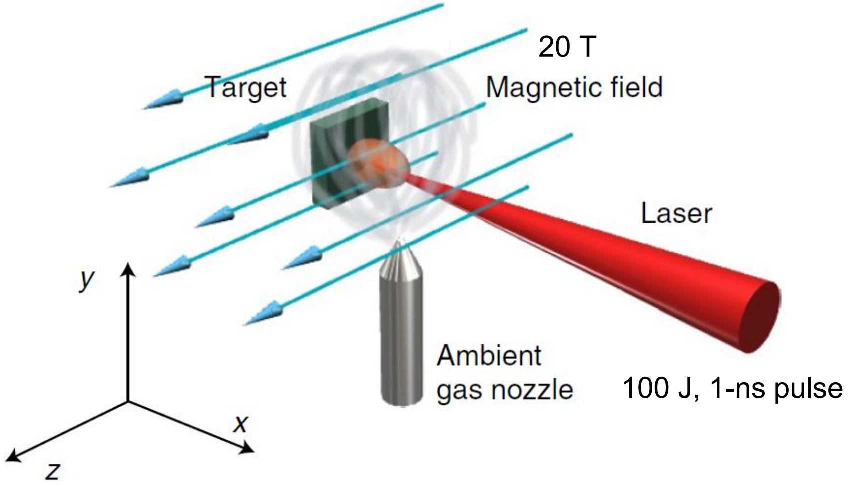
Strong density compression behind magnetic compression

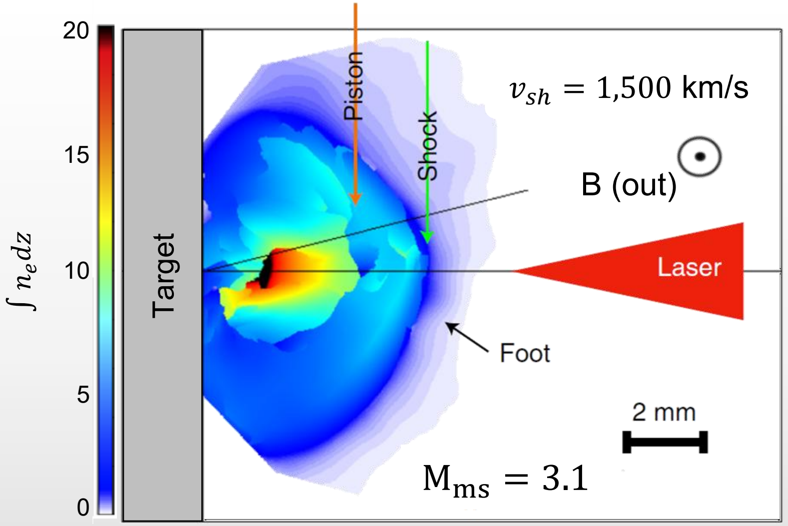
Significant adiabatic electron heating from compression

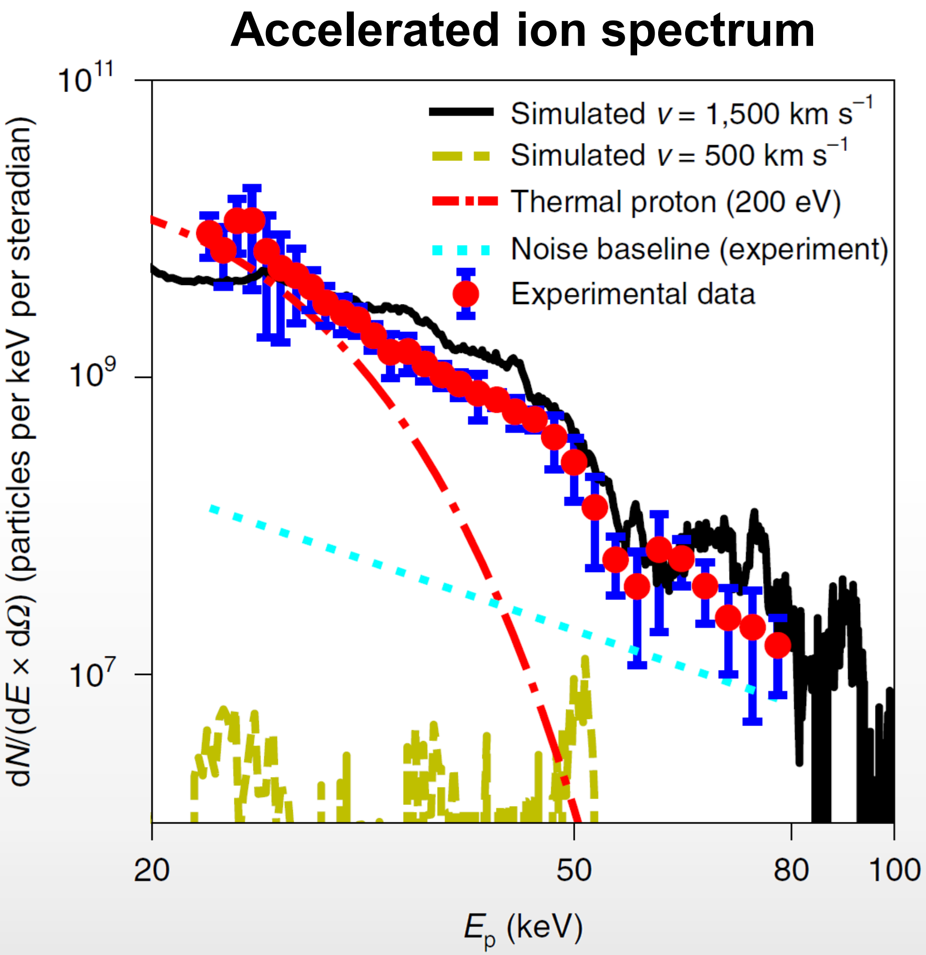
Piston-ambient coupling and heating, but downstream region has not fully formed

Schaeffer et al., Phys. Rev. Lett. (2017)
Schaeffer et al., Phys. Rev. Lett. (2019)

Accelerated particles have been created in these precursors





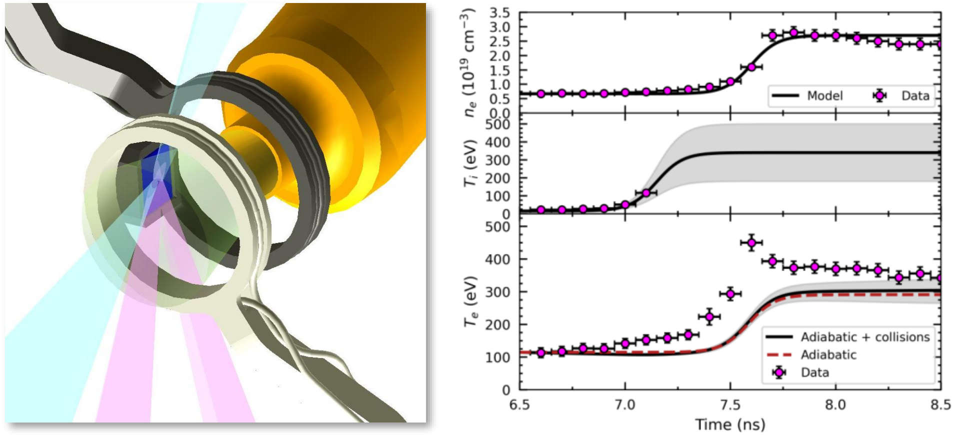


Yao et al., Nature Phys. (2021)

Summary

Part I. Anomalous electron heating

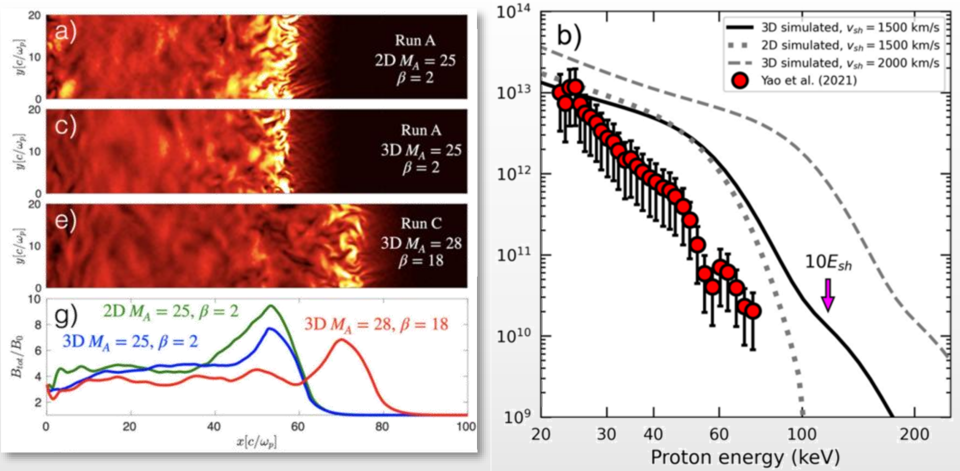
1. Fully-formed shocks on the OMEGA laser
2. Plasma parameters scanned as the shock streams through a stationary Thomson scattering collection volume
3. Analysis shows super-adiabatic electron heating through a foot and into the downstream, with non-Maxwellian downstream ions



[Valenzuela-Villaseca+arXiv:2509.12164 (in review at Phys. Rev. Lett.)]

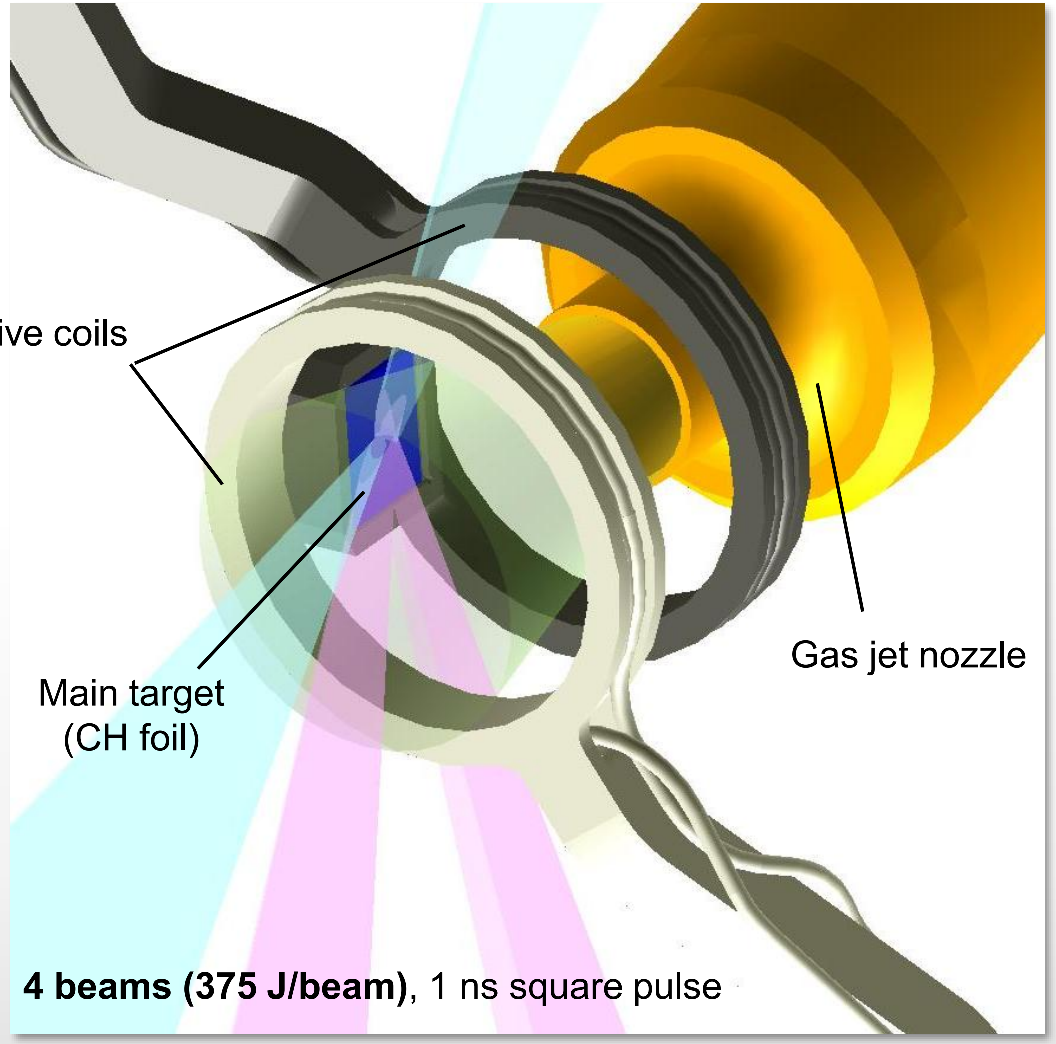
Part II. 3-D effects on ion acceleration

1. 3-D hybrid simulations (kinetic ions, fluid electrons) of perpendicular shocks
2. Strong ion acceleration for fast shocks (v_{sh} ~1,500 km/s)
3. Testing and validation of results by exploiting scaling relations

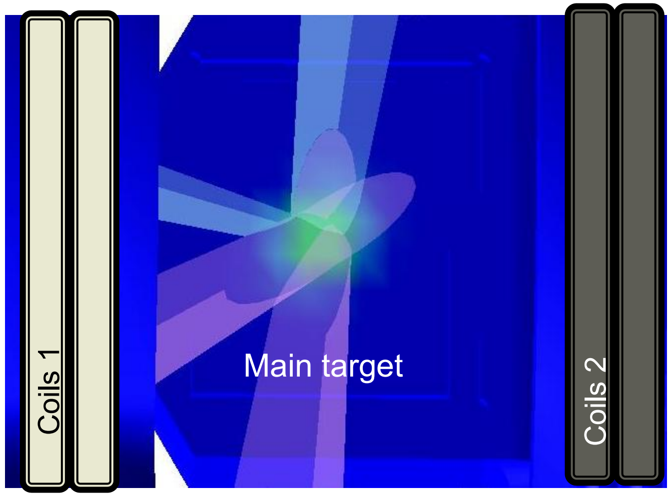


[Orusa & Valenzuela-Villaseca Phys. Plasmas (2025)]

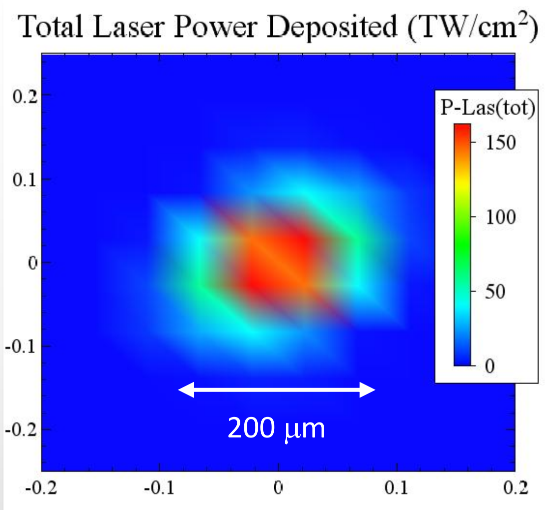
A new experimental platform to create and study long-lived collisionless shocks



Close up to target region



- 60 deg. incidence angle
- Each beam footprint is elongated

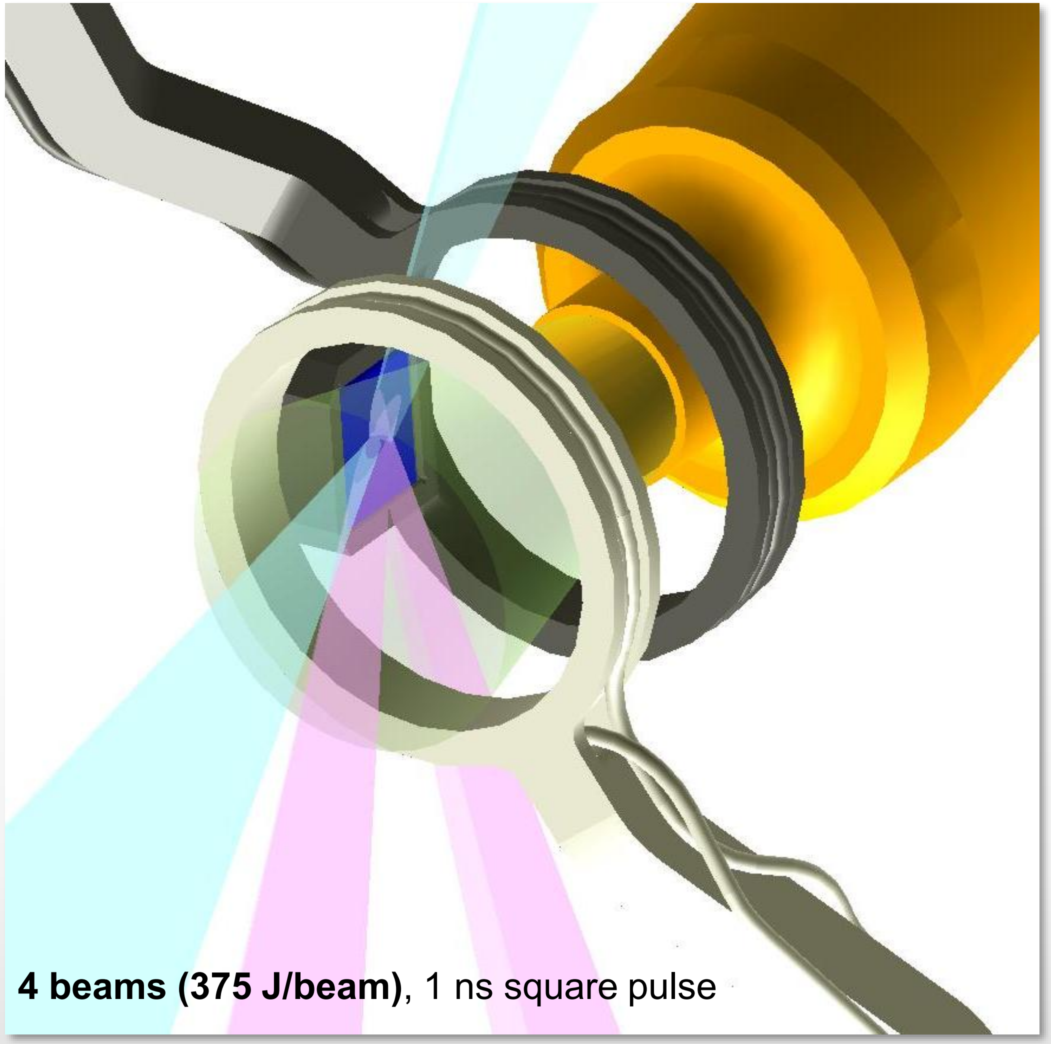


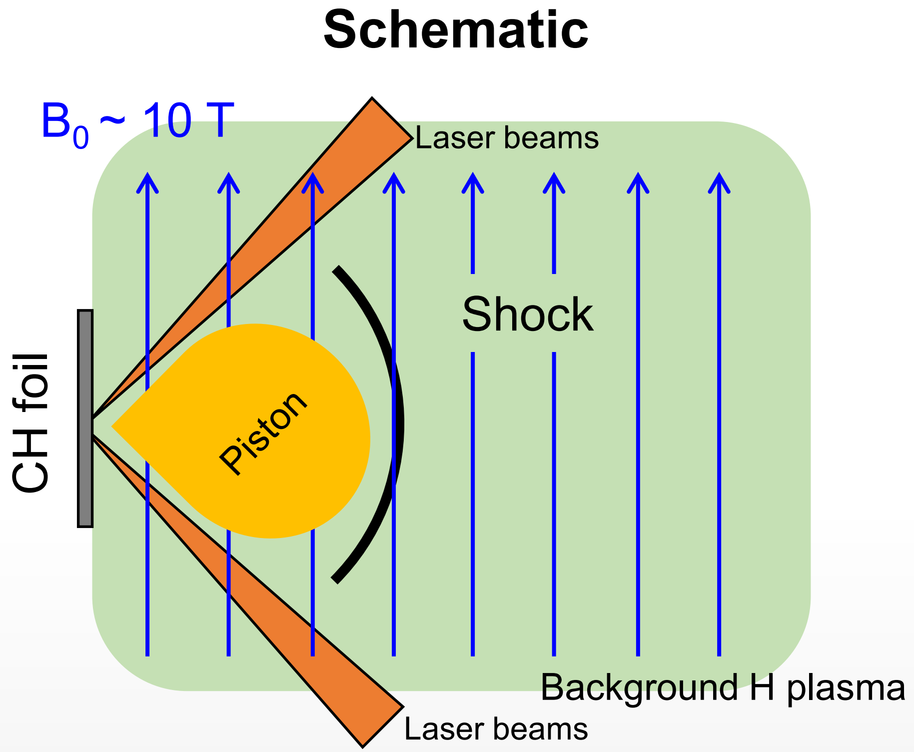
Simulated incident laser power

- Large irradiated region
- Quasi-axisymmetric spatial profile
- **Laser-target interaction releases an X-ray burst that pre-ionizes the hydrogen**

Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

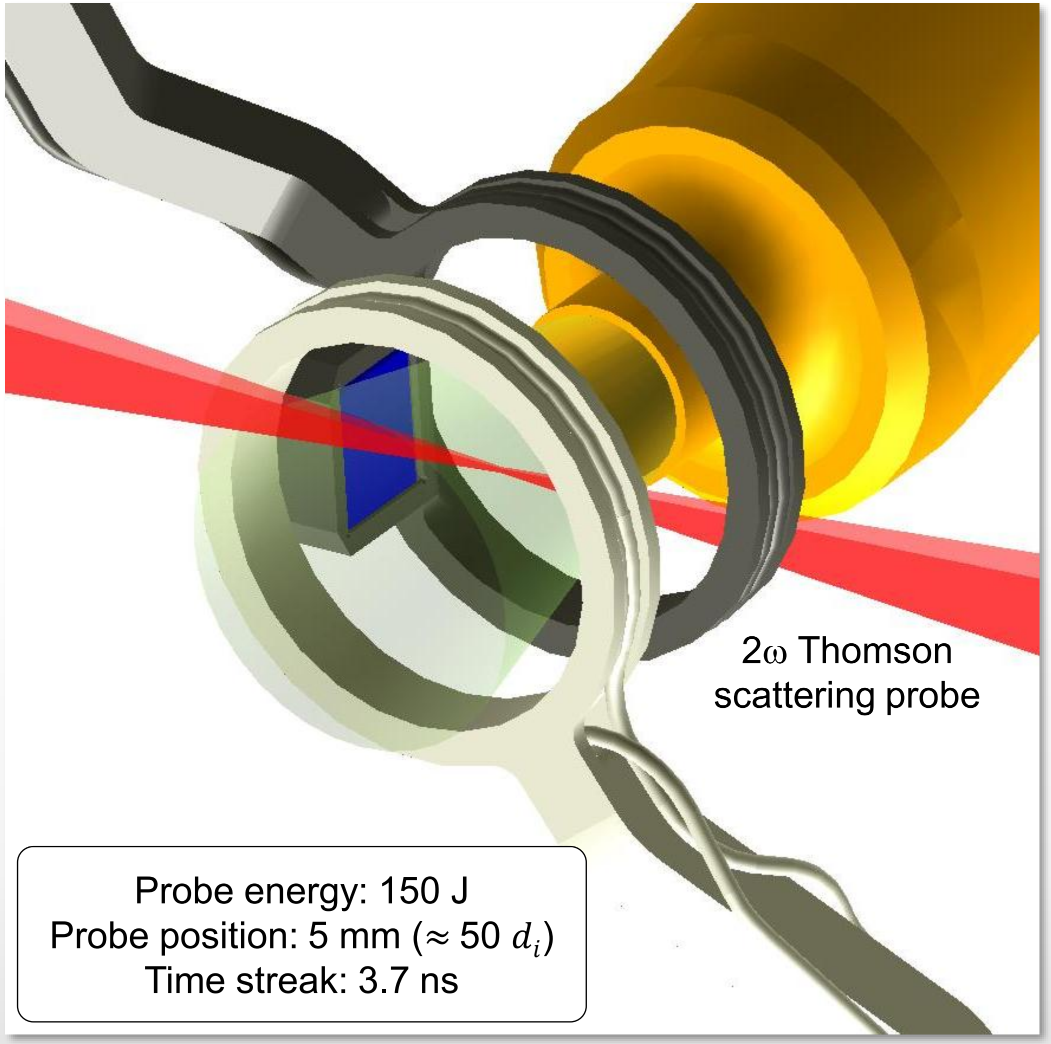
A new experimental platform to create and study long-lived collisionless shocks





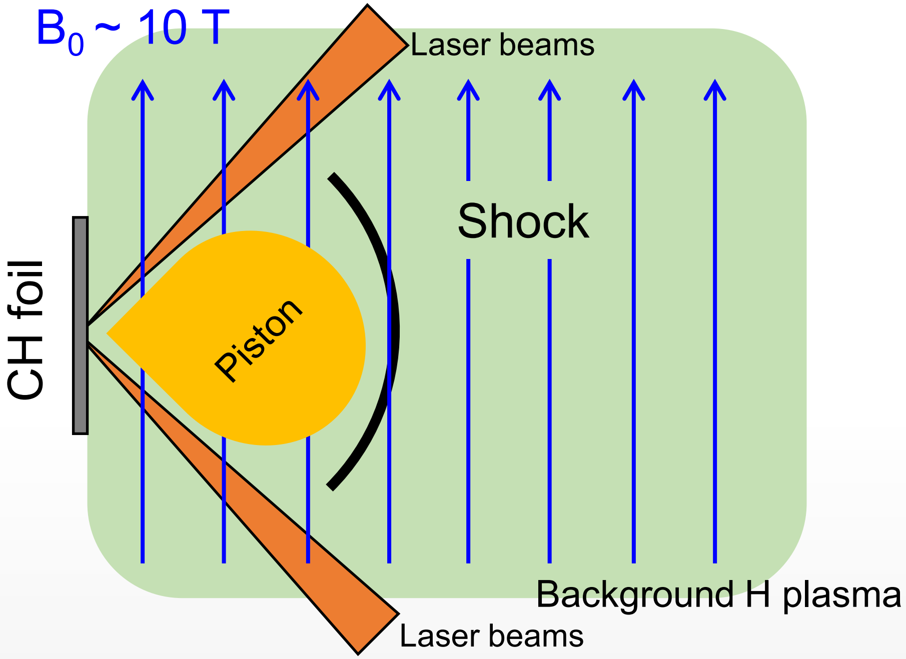
1. The region in front of a target is pre-magnetized
2. A gas jet fills the region with hydrogen gas
3. Lasers beams launch a hypersonic piston and X-rays ionize the gas

A new experimental platform to create and study long-lived collisionless shocks



Probe energy: 150 J
Probe position: 5 mm (≈ 50 d_i)
Time streak: 3.7 ns

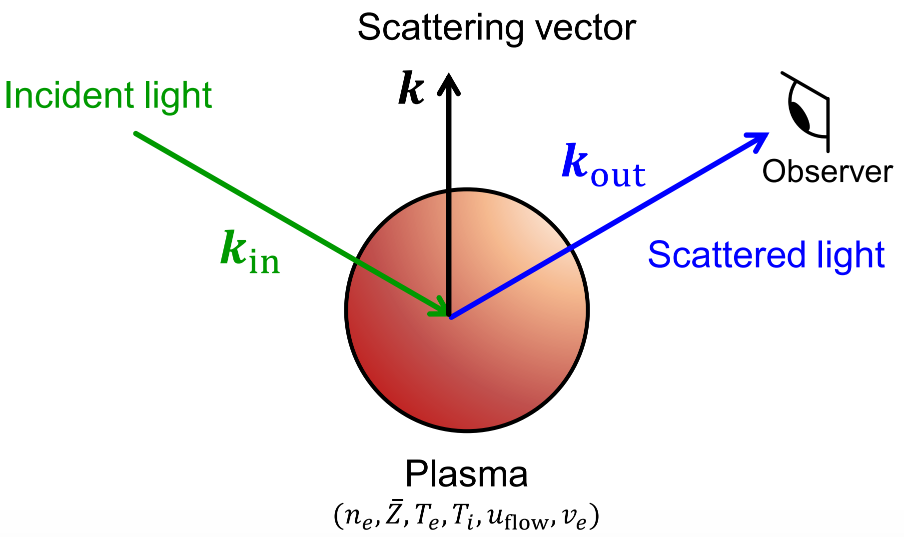
Schematic



1. The region in front of a target is pre-magnetized
2. A gas jet fills the region with hydrogen gas
3. Lasers beams launch a hypersonic piston and X-rays ionize the gas
4. A different laser beam probes the plasma through **Optical Thomson Scattering (OTS)**

Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

The plasma was probed using Optical Thomson Scattering (OTS) in the collective regime



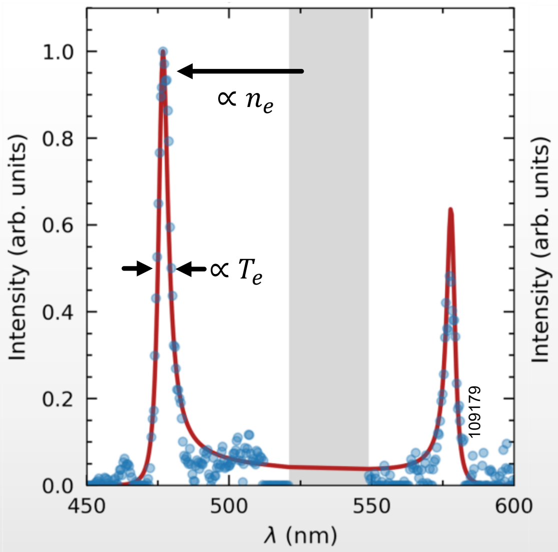
Thomson scattering is an elastic process

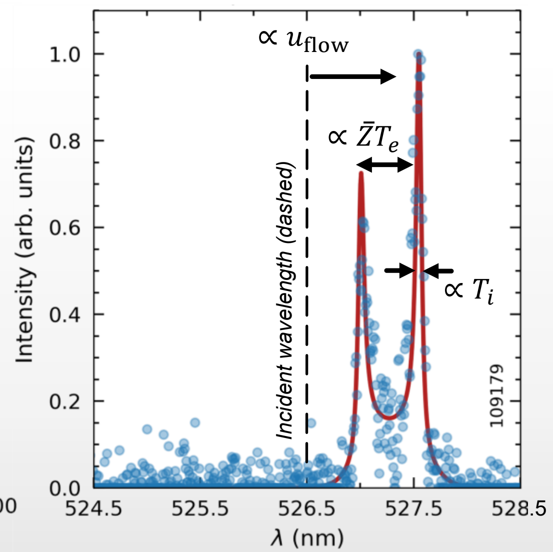
$$\underbrace{\omega_{\text{out}} = \omega_{\text{in}} + \omega}_{\textit{Energy conservation}}; \quad \underbrace{\boldsymbol{k}_{\text{out}} = \boldsymbol{k}_{\text{in}} + \boldsymbol{k}}_{\textit{Momentum conservation}}$$

The diagnostic is sensitive to particle motion in the direction of the scattering vector $\boldsymbol{k} = \boldsymbol{k}_{\text{out}} - \boldsymbol{k}_{\text{in}}$

In a plasma, the scattered spectra is shaped by the dynamic form factor

$$S(k,\omega) = \underbrace{\frac{2\pi}{k}\left|1 - \frac{\chi_e}{\epsilon}\right|^2 f_{e0}\left(\frac{\omega}{k}\right)}_{\textbf{Electron plasma wave (EPW)}} + \underbrace{\frac{2\pi}{k}\sum_s \frac{\bar{Z} n_s}{n_e}\left|\frac{\chi_e}{\epsilon}\right|^2 f_{s0}\left(\frac{\omega}{k}\right)}_{\textbf{Ion-acoustic wave (IAW)}}$$





Froula, et al., *Plasma scattering of electromagnetic radiation*, Elsevier (2011)
Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

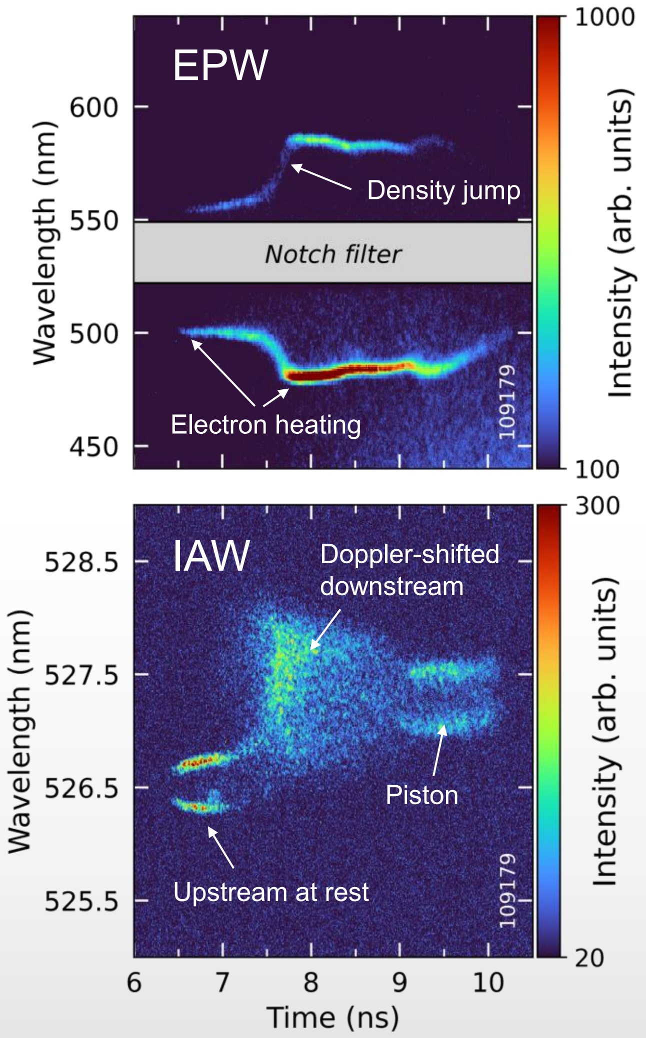
As the shock passes through the collection volume, we scan from the upstream to upstream

Electron-plasma wave

- Resonance separation: **density jump at 7.5 ns**
- Resonance broadening**: electron heating** correlated with density jump

Ion-acoustic wave

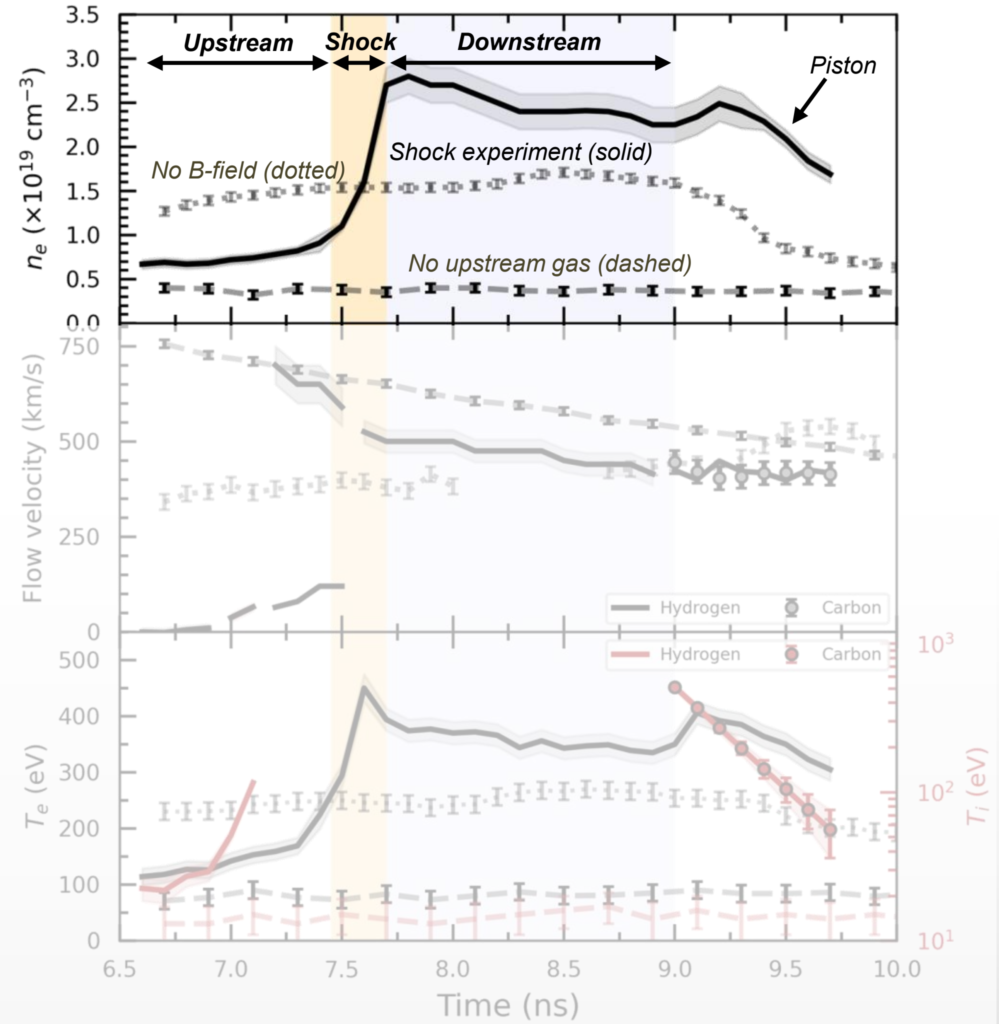
- Initially, no Doppler shift of spectrum: upstream at rest
- Overall shift: **ion flow velocity – accelerated downstream**
- **Broadening of ion feature (not consistent with Maxwellian VDF**): hard to analyse!
- At late times, carbon and hydrogen are fully mixed, indicating the measurement of the piston



Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

We observe fully formed shocks and rule out significant collisional, hydrodynamic coupling

- Density jump consistent with **fully formed collisionless shock** $(R = 4)$
- $\tau_{\mathrm{ramp}} \approx \omega_{ci}^{-1} \sim 4d_i$ in the compressed field
- **No B-field = no density jump:** no collisional, hydrodynamic coupling
- **No upstream = no density jump**



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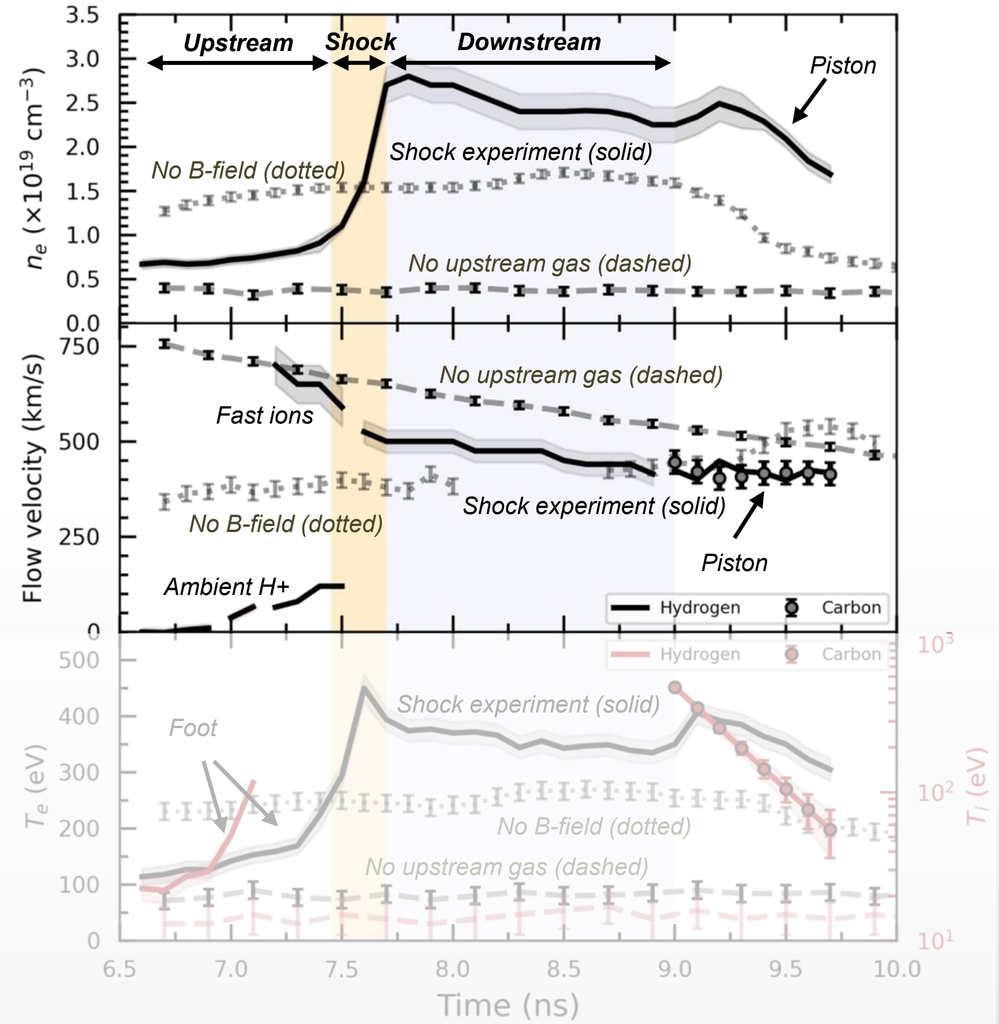
- **Fast ion population streaming ahead of the shock** (reflected ions?)
- Downstream flow speed = 500 km/s (consistent with jump conditions)
- **No B-field = no velocity drop**



Plasma parameters

$B^{(u)}$ = 10.5 T
$n_e^{(u)}$ = 5e18 cm-3
$u_{\text{flow}}^{(d)}$ = 500 km/s
v_{sh} = 670 km/s

Dimensionless parameters

$\lambda_{\text{mfp}}^{i\backslash i}$ = 2 cm >> 500 μm = L_n
M_A = 6.5 (=M_S)
M_{ms} = 9
$\beta \approx 1$

Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

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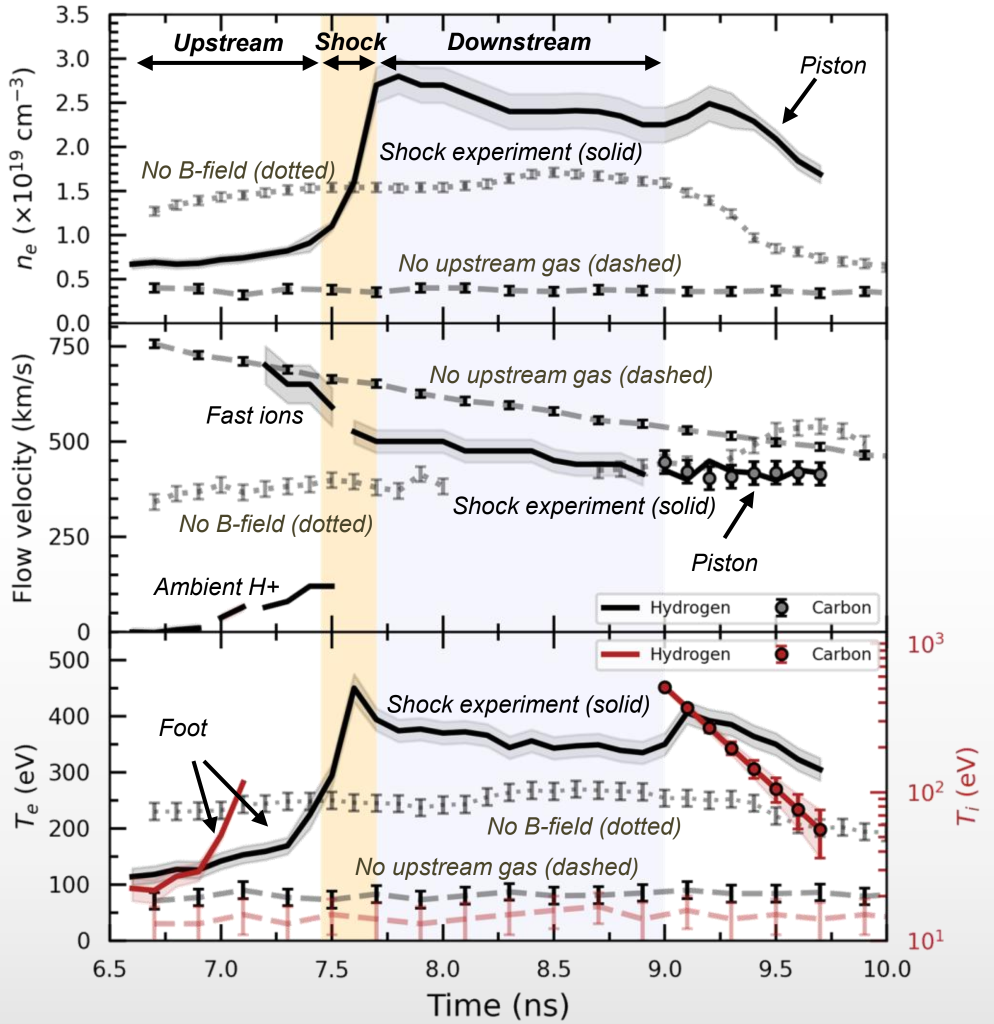
- **Fast ion population streaming ahead of the shock** (reflected ions?)
- Downstream flow speed = 500 km/s (consistent with jump conditions)
- **No B-field = no velocity drop**

- **Strong electron heating:** $T_e^{(u)} = 115 \pm 15 \text{ eV} \rightarrow T_e^{(d)} = 390 \pm 20 \text{ eV}$
- **Super-adiabatic:** $T_e^{(d)}/T_e^{(u)} = 3.5 \pm 0.5 > R^{2/3} = 2.5 \pm 0.1$
- **Foot formation:** electron and ion heating ahead of the shock
- No B-field, no upstream = no temperature jump

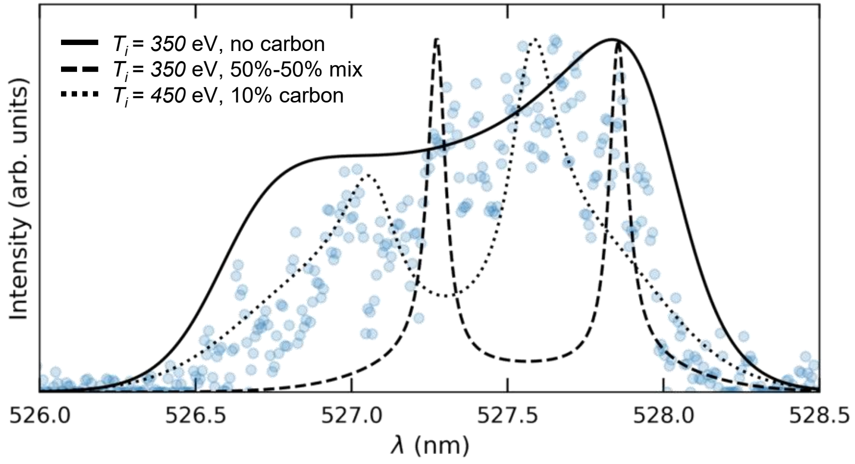


Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

Even if no direct fit can be done, we can infer the ion temperature downstream of the shock

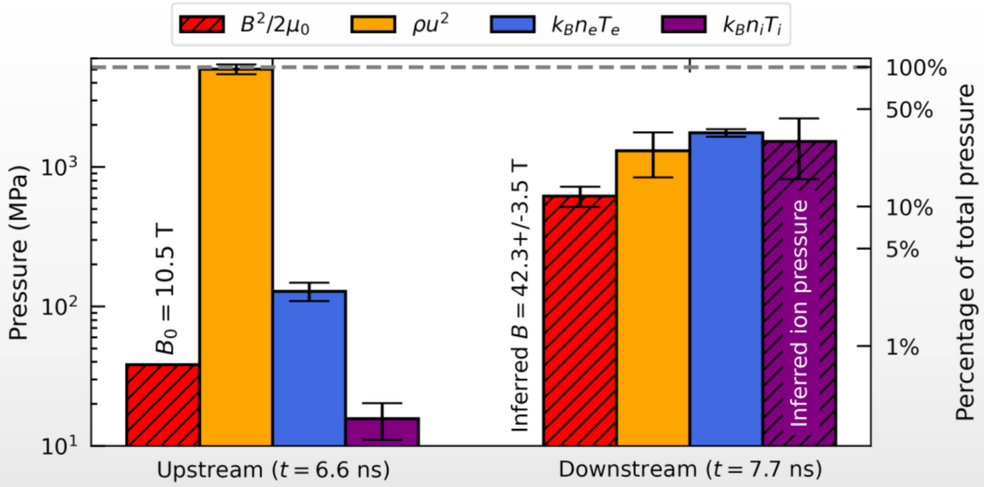
Ion-acoustic wave spectrum

- Downstream IAW broadening consistent with T_i ~ few hundred eV
- Precise shape not adequately described by a Maxwellian VDF



Pressure balance

- Upstream strongly dominated by ram pressure
- Downstream has scrambled energetics
- Downstream magnetic field calculated from flux compression $B^{(d)} = RB^{(u)}$: always subdominant
- Ion pressure = 30% deficit from which ion temperature can be inferred



Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

The inferred ion temperature is insensitive to ion abundances this plasma regime

We parameterize the pressure balance equation as a function of hydrogen and carbon fractions f_H, f_C

Total pressure

$$p_{tot} = p_{\mathrm{th}}(f_H, f_C) + p_{\mathrm{ram}}(f_H, f_C) + p_M = \text{const.}; \quad f_H + f_C = 1$$

Thermal pressure

$$p_{\mathrm{th}} = n_e k_B (T_e + T_i/\bar{Z})$$

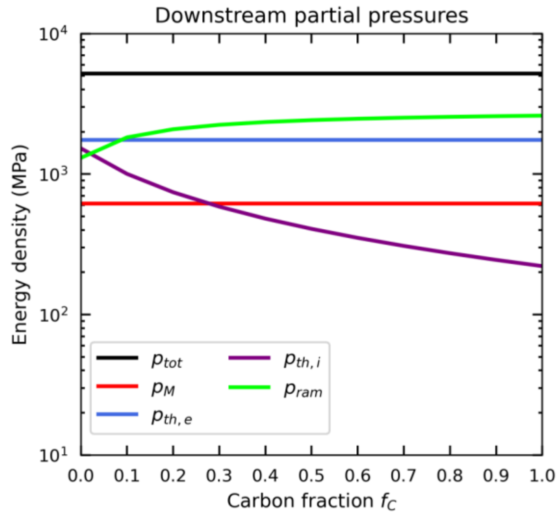
Ram pressure

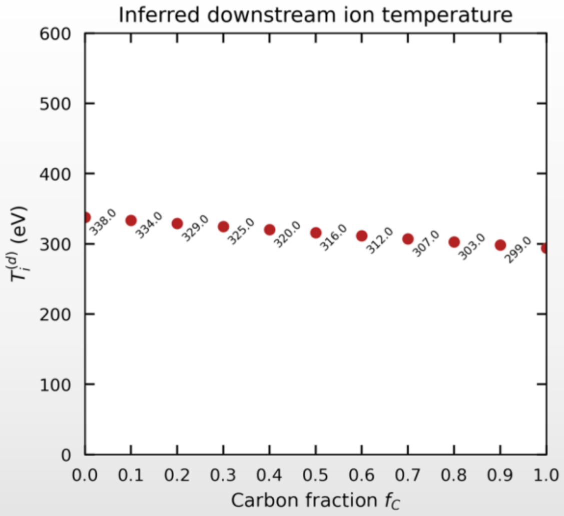
$$p_{\mathrm{ram}} = n_e \left(\sum_{\beta=H,C} \frac{f_\beta m_\beta}{\bar{Z}} \right) u_{\mathrm{flow}}^2$$

Average charge state

$$\bar{Z} = \sum_{\beta=H,C} f_\beta Z_\beta$$

Result: downstream T_i = 350±160 eV, with small dependance on carbon abundance





Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

Electron-ion collisions are insufficient to explain the electron heating, and it is therefore collisionless

- Electron heating equation

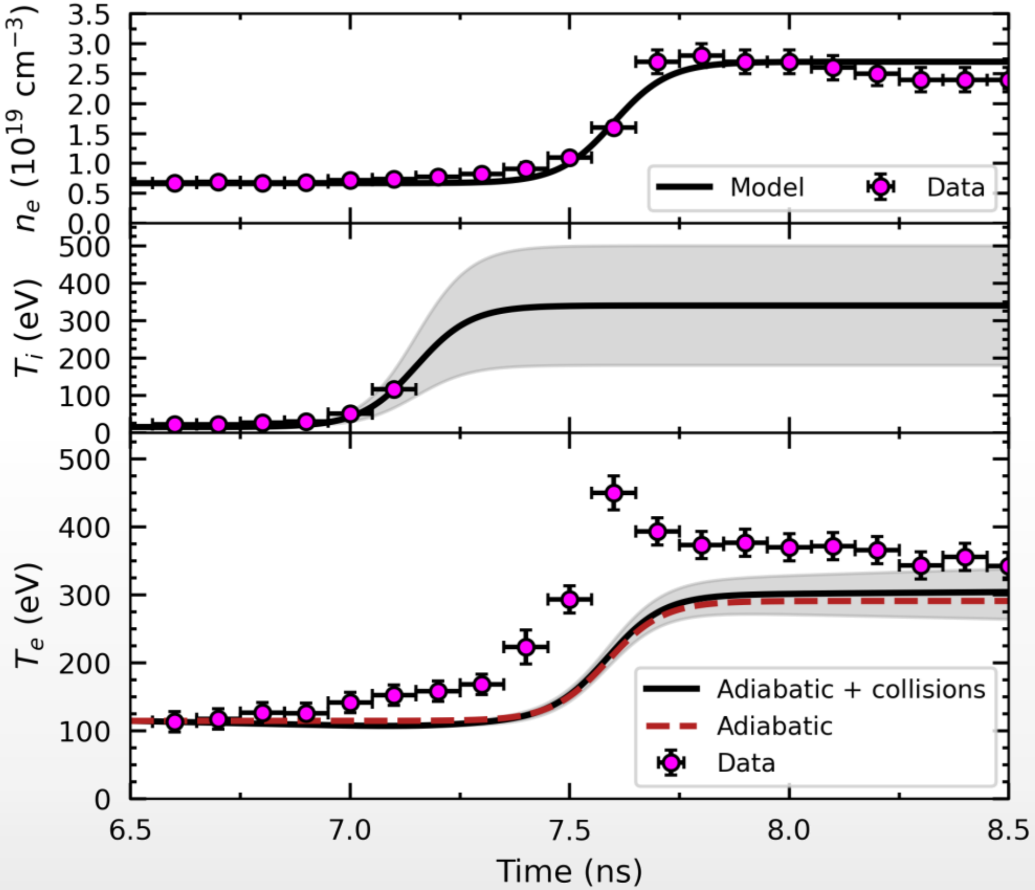
$$n_e \frac{dT_e}{dt} = \frac{2}{3} T_e \frac{dn_e}{dt} + \sum_\beta \nu_{\text{eq}}^{e\backslash i}(T_i - T_e)$$

- Estimated downstream electron temperature

$$T_e^{(d)} = \underbrace{\left(\frac{n_e^{(d)}}{n_e^{(u)}}\right)^{2/3} T_e^{(u)}}_{\approx 290 \text{ eV}} + \underbrace{\nu_{\text{eq}}^{e\backslash\text{i}} \tau_{\text{ramp}} \left(T_i^{(d)} - T_e^{(u)}\right)}_{\approx 5 \text{ eV}} \approx 300 \text{ eV}$$

- Numerical calculation

1. Super-adiabatic electron heating in the foot
2. Sustained anomalous heating in the downstream



Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

Summary of Part I.

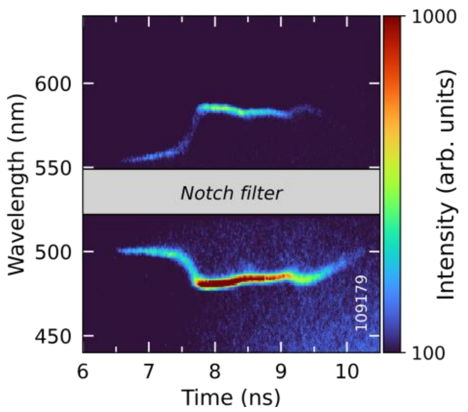
1. New platform to study **fully-formed magnetized shocks** in the laboratory

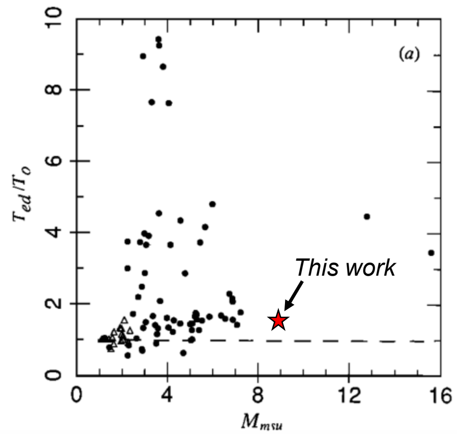
2. Significant **super-adiabatic electron heating**

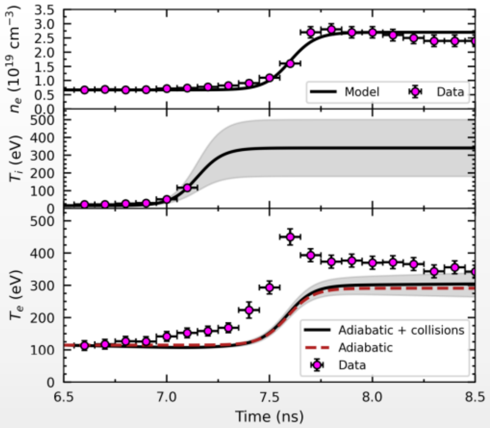
3. Significant ion heating inferred from jump conditions and consistent with OTS broadening

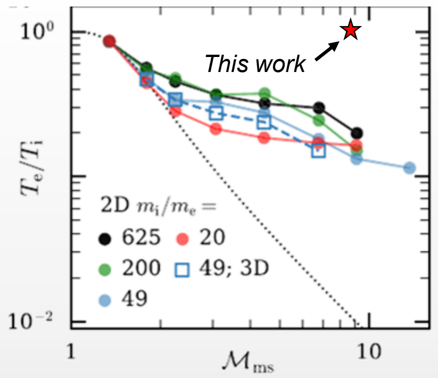
4. Downstream T_e/T_i = 1.2 ± 0.6, which **excludes numerical and theoretical trends**

5. Results in line with SNR observations and spacecraft measurements around the Earth's bow-shock







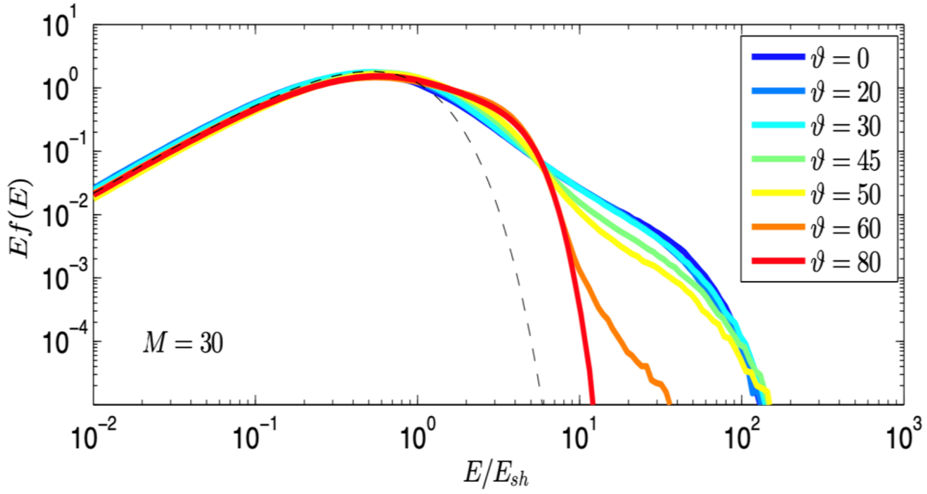


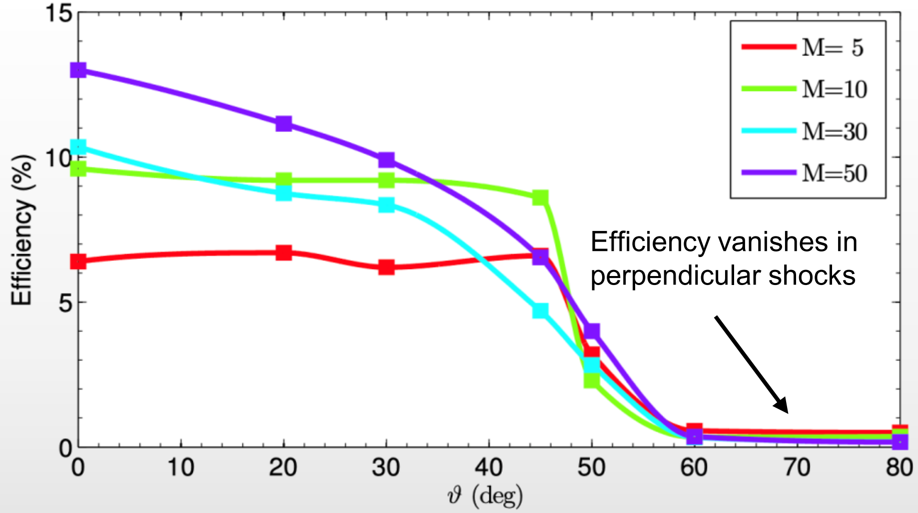
This project supported by DOE/NNSA Awards No. DE-NA0004033, DE-NA0003856, and DE-NA0004144

Valenzuela-Villaseca et al., arXiv:2509.12164 (in review at Phys. Rev. Lett.)

It has been understood, largely through 1-D and 2-D simulations, that perpendicular shocks do not produce energetic ions

- **Shock drift acceleration:** ions are accelerated when in contact with the shock by $v_{sh} \times B$ electric field
- **Diffuse shock acceleration:** ions are accelerated by repeated crossings through the shock into the upstream and back
- **In 2-D perpendicular shocks, magnetic fields act as walls**, decreasing acceleration efficient

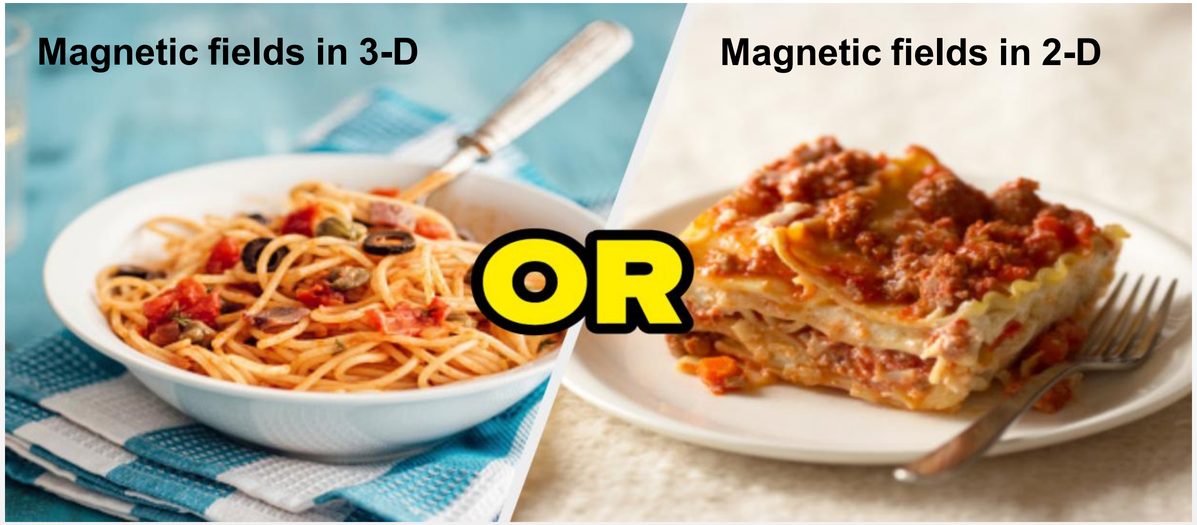




Caprioli & Spitkovsky, Astroph. J. (2014)

However, in 3-D and at large enough M_A, ions can leak through the magnetic turbulence back into the upstream

- In **2-D**, magnetic turbulence provides *corrugation:* no cross-field ion diffusion
- In **3-D**, magnetic turbulence provides *porosity:* cross-field ion diffusion allowed



Orusa & Caprioli, Phys. Rev. Lett (2023)
Orusa et al., (arXiv:2507.1343 in review at Astrophys. J.)
Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

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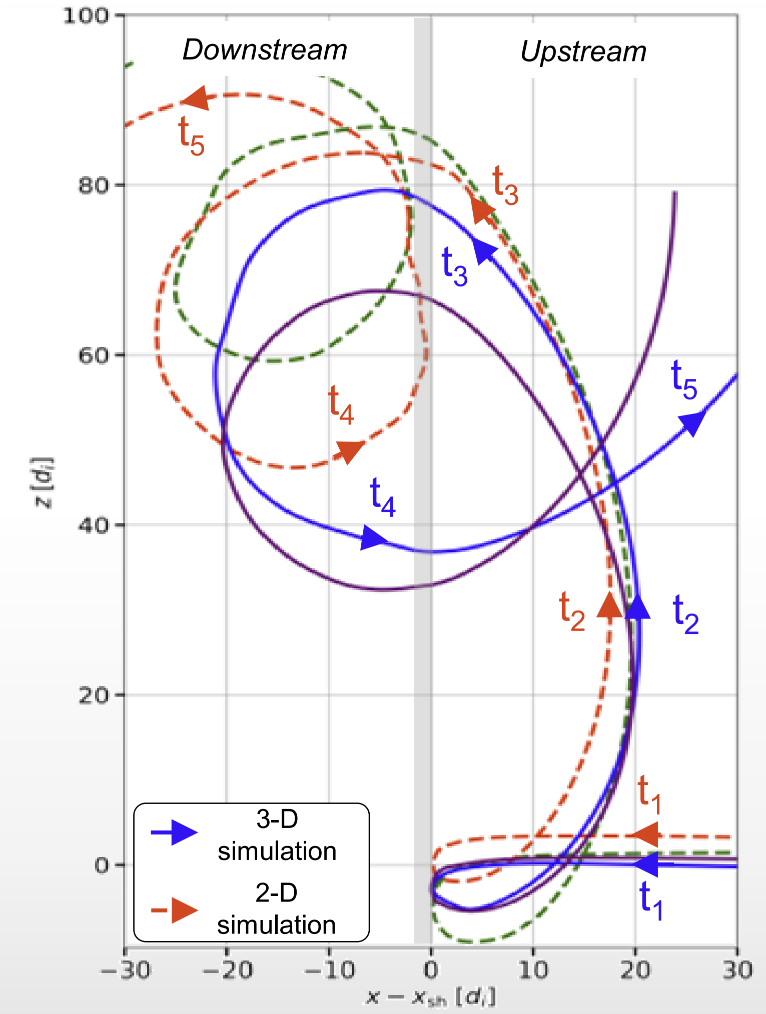
Requirements for 3-D ion acceleration

- High M_A (≥ 20), magnetic fluctuations must be large enough
- High M_S (≥ 13), implies $\beta \sim 1$ at high M_A
- Ion occurs after ~ 10 ion gyrations: fast for astrophysics, but slow for experiments!

Laboratory experiments are supported by 1-D or 2-D simulations: 1) are they missing something? 2) can we test these results in the laboratory?



Orusa & Caprioli, Phys. Rev. Lett (2023)
Orusa et al., (arXiv:2507.1343 in review at Astrophys. J.)
Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

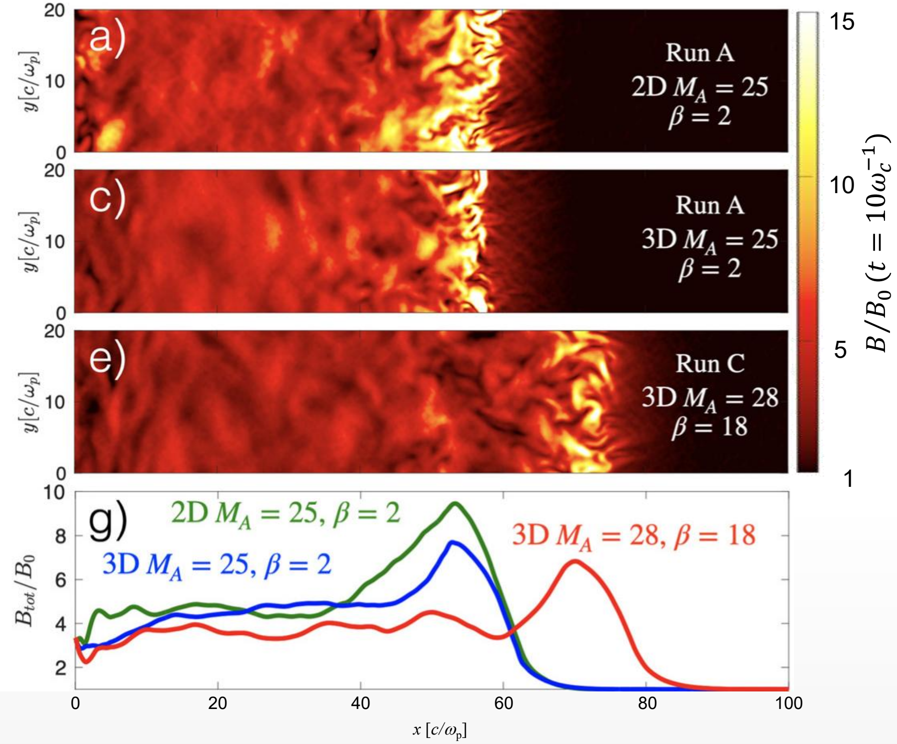
We used hybrid-PIC simulations to study 3-D ion acceleration

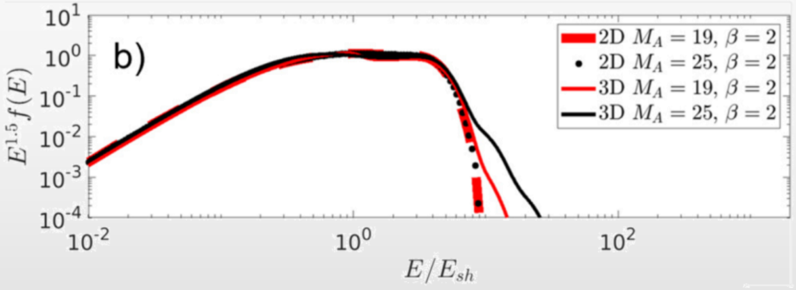
- Simulations done in long boxes, but at least 14 di in the transverse direction
- Scan over values of M_A = 19, 25, 28 and β = 2, 5, 18

Run	M_A	β	M_s	$\varepsilon\,(>10E_{sh})$	R	α
A	25	2	19	0.3%	4.2	5.4
B	25	5	13	0.2%	4.3	5.7
C	28	18	7	0.05%	3.5	8
D	19	2	15	0.04%	4.3	8

Super-thermal tail (A, B)

No super-thermal tail (C, D)



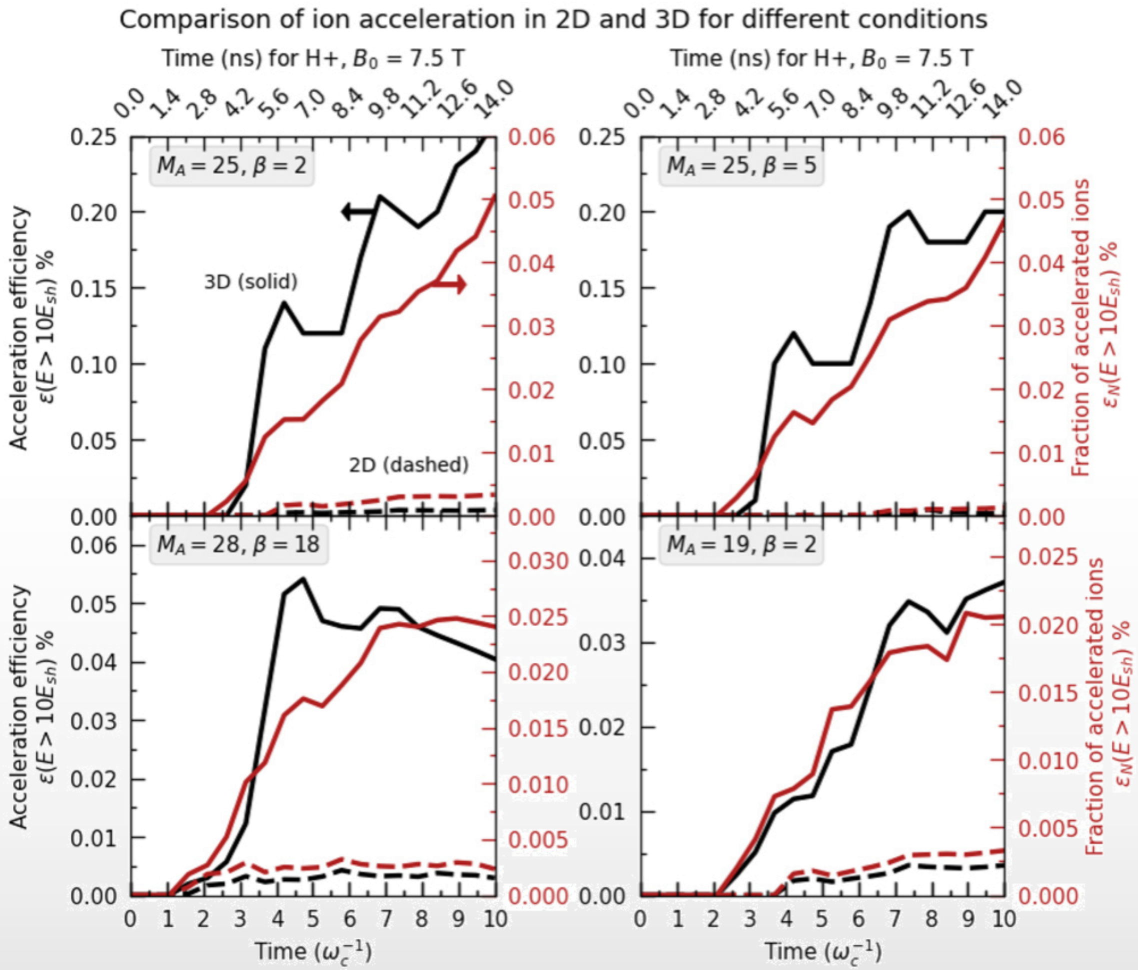


At 10 ion gyrations, thresholds are M_A^* = 25 and β^* = 5 (M_S^* = 13) for super-thermal tails to develop

Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

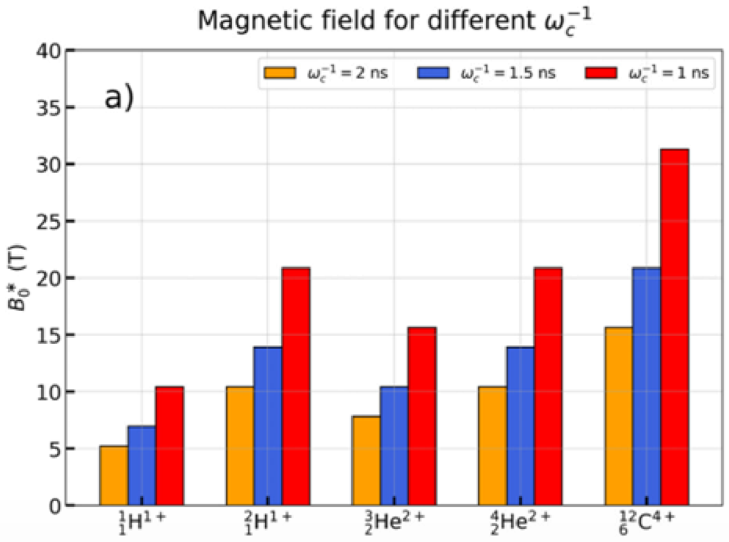
Simulations differences on the acceleration efficiency at early times (< 10 ns) at laboratory magnetic fields

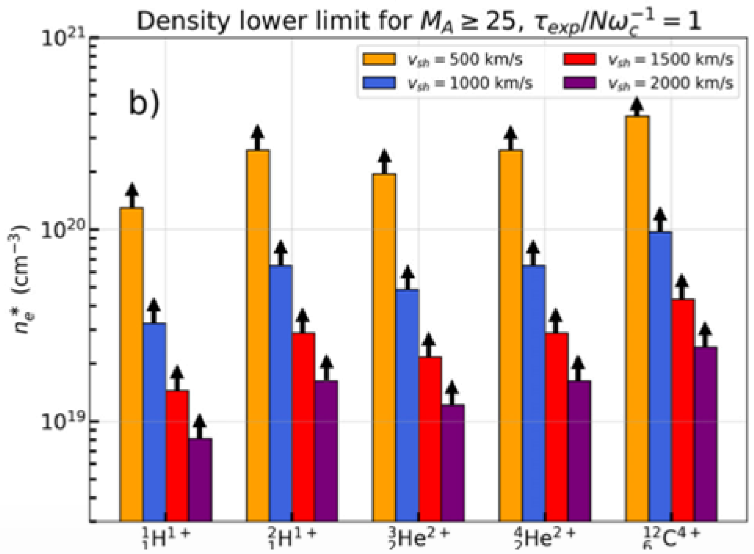
- At high β = 18, acceleration stalls after 7 ion gyrations at 0.025% of ion fraction
- At low β = 2, acceleration continues beyond 0.06% after 10 ion gyrations

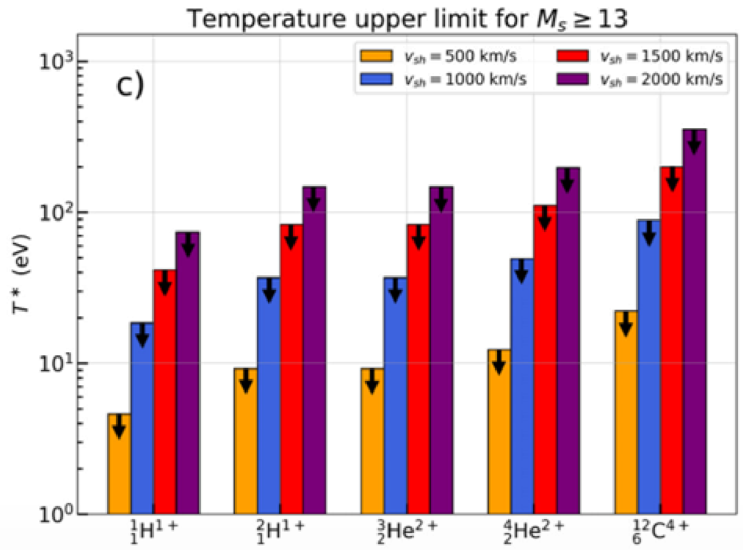


Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

Threshold scaling could allow to turn 3-D on and off by selecting the ion species, density, and/or temperature upstream of the shock







$$B_0 = \left(\frac{A}{Z}\right)\left(\frac{m_p}{e}\right)\frac{N_{\mathrm{crit}}}{\tau_{\mathrm{exp}}} \approx 10.4\left(\frac{A}{Z}\right)\left(\frac{10\,\mathrm{ns}}{\tau_{\mathrm{exp}}}\right)\mathrm{T},$$

$$n_e^* = \left(\frac{A}{Z}\right)\left(\frac{m_p}{\mu_0 e^2}\right)\left(\frac{N_{\mathrm{crit}}}{\tau_{\mathrm{exp}}}\right)^2\left(\frac{M_{A,\,\mathrm{crit}}}{v_{sh}}\right)^2$$
$$\approx 35\times 10^{18}\left(\frac{A}{Z}\right)\left(\frac{10\,\mathrm{ns}}{\tau_{\mathrm{exp}}}\right)^2\left(\frac{1000\,\mathrm{km/s}}{v_{sh}}\right)^2 \mathrm{cm}^{-3},$$

$$T^* = \left(\frac{A}{1+Z}\right)\left(\frac{m_p}{\gamma k_B}\right)\left(\frac{v_{sh}}{M_{s,\,\mathrm{crit}}}\right)^2,$$
$$\approx 50\left(\frac{2A}{1+Z}\right)\left(\frac{v_{sh}}{1650\,\mathrm{km/s}}\right)^2 \mathrm{eV},$$

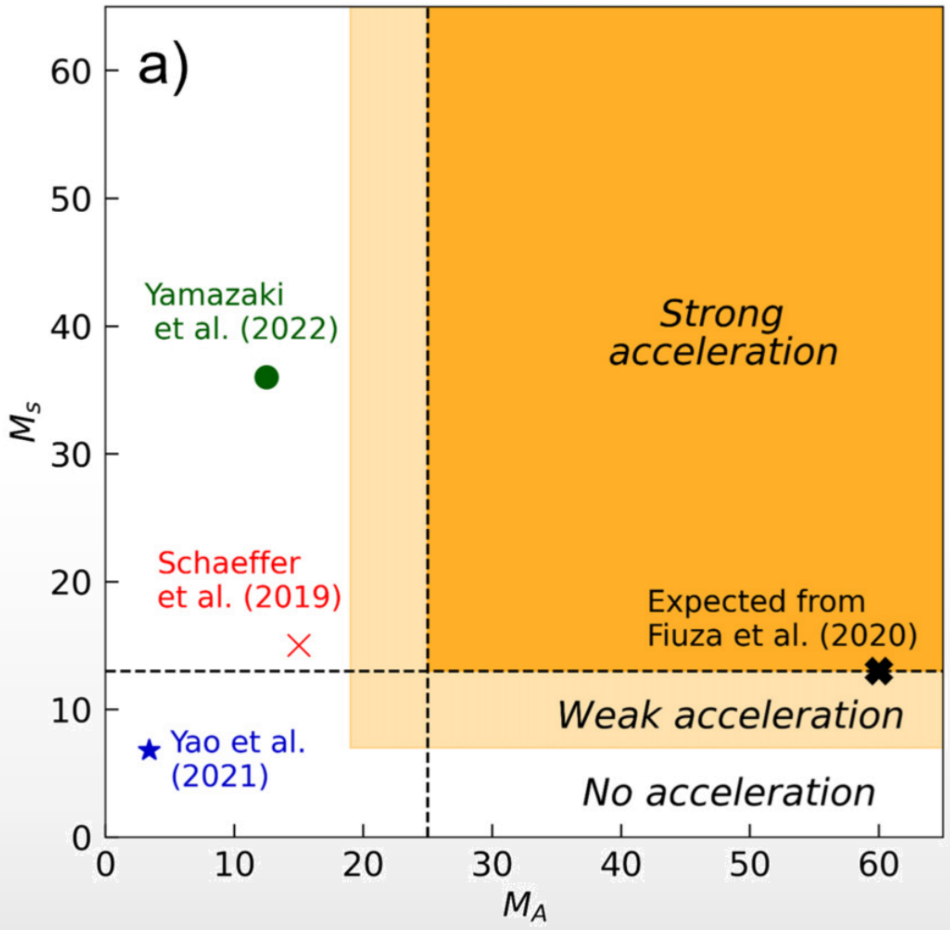
- Criterion 1: at least 10 ion gyrations
- Magnetic field sets gyration time

- Criterion 2: M_A > 25
- At fixed B-field, density sets M_A

- Criterion 3: M_S > 13
- At a given shock speed, temperature sets M_S

We do not expect 3-D acceleration in previous experiments, and 1-D and 2-D simulations should be sufficient for their modelling

We don't expect previous experiments to produce 3-D ion acceleration, but they are close!

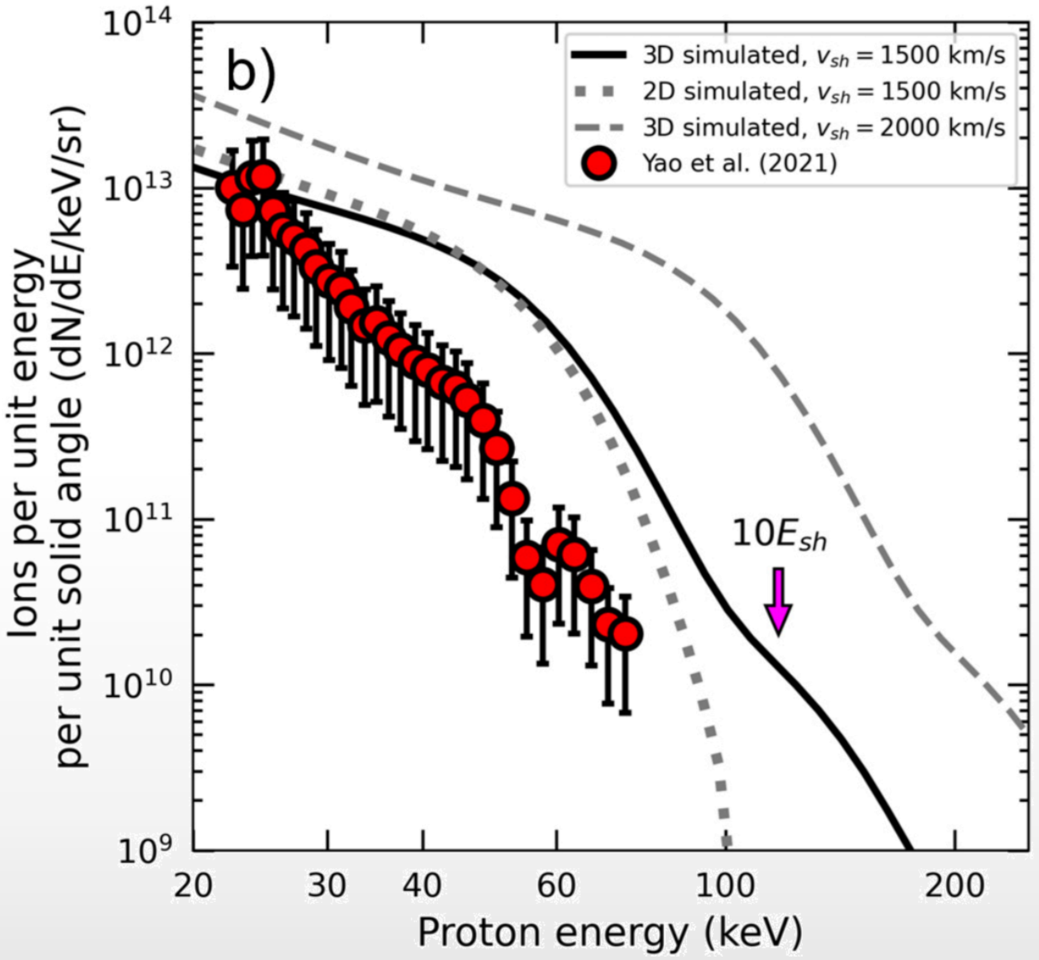


Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

Even though the ion acceleration is small ($\varepsilon \approx 0.1\%$), laboratory instruments are sensitive enough to detect them

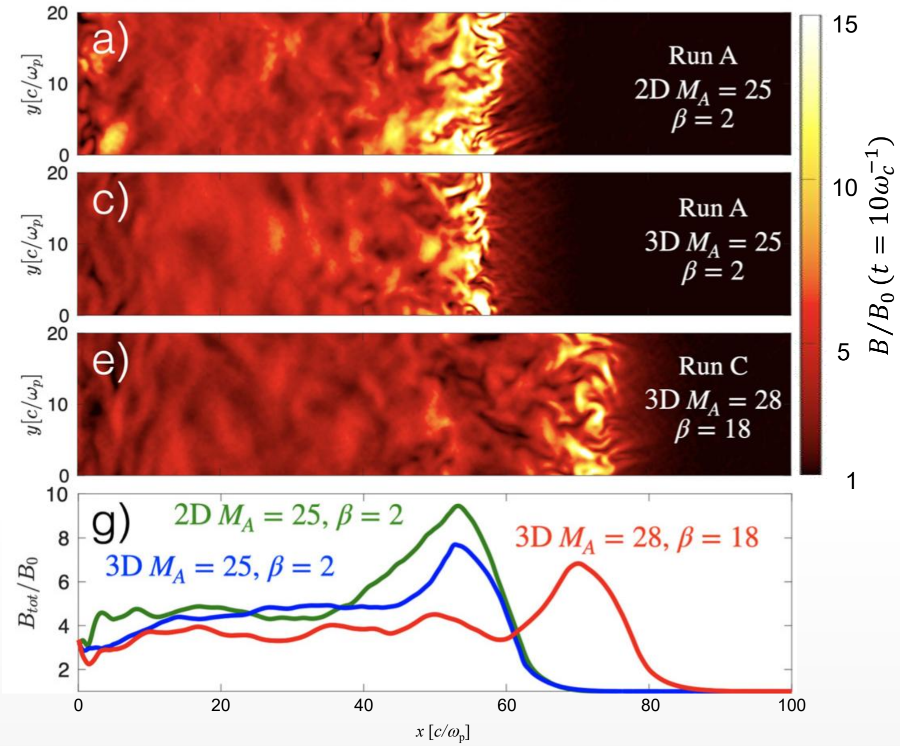
- Given the simulated ion fractions and super-thermal tails, we can calculate the ion spectrum for more optimistic conditions
- We used calibration factors from experiments at LULI (France)

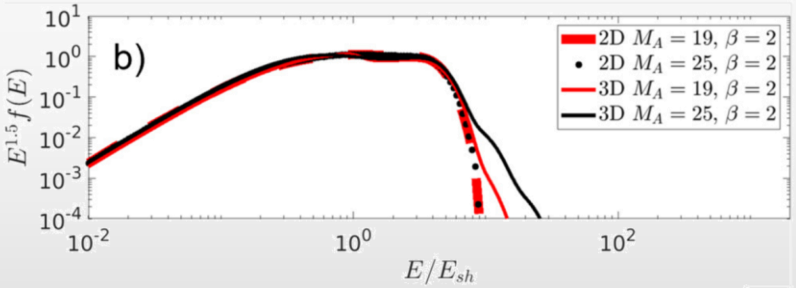
Current instruments *should* be able to measure super-thermal tails!



Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)
Yao et al., Nature Phys (2021)

Summary of Part II.





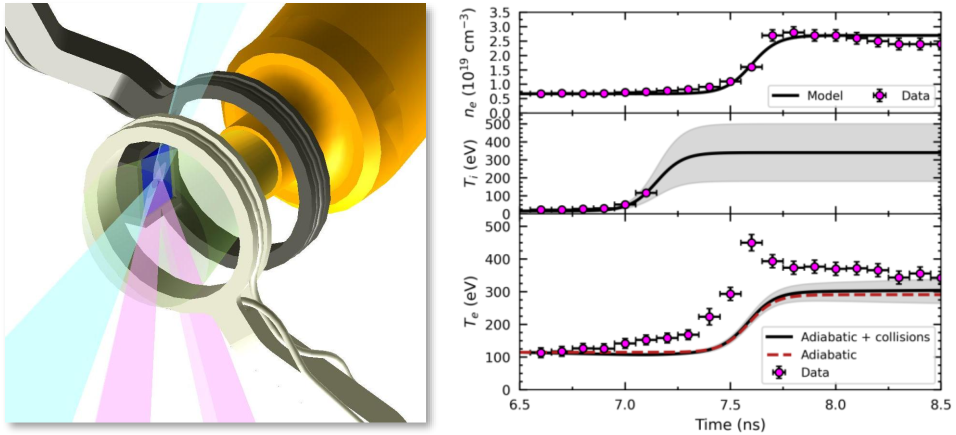
1. Hybrid simulations showed that 3-D ion acceleration can be achieved and measured under laboratory conditions
2. Previous experiments should not produce significant ion acceleration
3. New laboratory experiments could allow to benchmark astrophysical codes directly

Orusa & Valenzuela-Villaseca, Phys. Plasmas (2025)

Summary

Part I. Anomalous electron heating

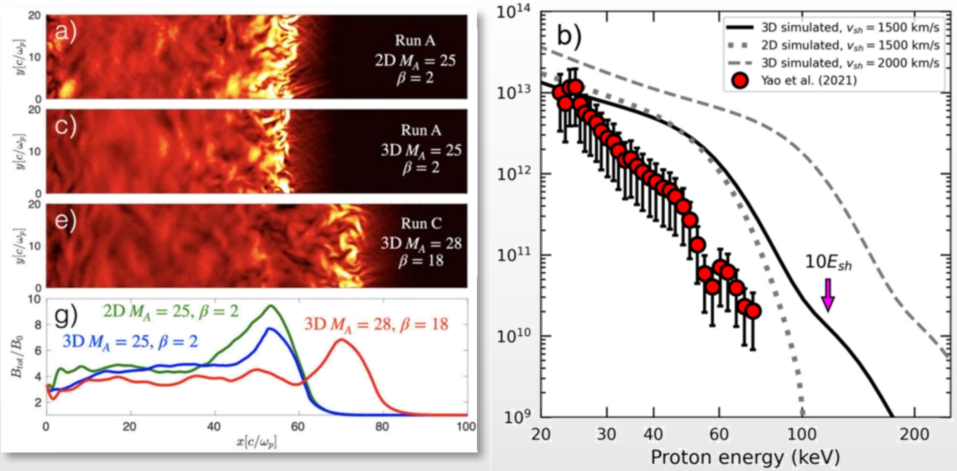
1. Fully-formed shocks on the OMEGA laser
2. Plasma parameters scanned as the shock streams through a stationary Thomson scattering collection volume
3. Analysis shows super-adiabatic electron heating through a foot and into the downstream, with non-Maxwellian downstream ions



[Valenzuela-Villaseca+arXiv:2509.12164 (in review at Phys. Rev. Lett.)]

Part II. 3-D effects on ion acceleration

1. 3-D hybrid simulations (kinetic ions, fluid electrons) of perpendicular shocks
2. Strong ion acceleration for fast shocks (v_{sh} ~1,500 km/s)
3. Testing and validation of results by exploiting scaling relations



[Orusa & Valenzuela-Villaseca Phys. Plasmas (2025)]